This material is neither intended to be distributed to Mainland China investors nor to provide securities investment consultancy services within the territory of Mainland China. This material or any portion hereof may not be reprinted, sold or redistributed without the written consent of J.P. Morgan.

Delta-One Flows & Positioning

Record Semis Inflows With Pivot to Unlevered; Munis/ Mid-Term USTs Bid, Utilities/Money Market/Gold Sold; CTAs De-Risk US/Japan/Brazil

In this publication, we analyze flow and positioning metrics for global futures and US-listed ETFs across asset classes.

Weekly Highlights

ETF investors aggressively bought the Semis dip again as Semiconductor ETFs logged record inflows. Regionally, Canada stood out with strong inflows, while Brazil was quiet in the final week ahead of the runoff election. Non-tech sectors were weak, led by Utilities. In Fixed Income, Munis stayed firmly bid and intermediate Treasuries saw renewed demand, while Money Market ETFs flipped to outflows. Cross-asset hedges were pared back: gold was sold and managed-futures ETFs saw material outflows. In futures, equities and rates both saw ~1z net buying, while leveraged funds covered RTY shorts but added to Ultra Bond shorts. CTA signals point to de-risking in US and Japan equities as volatility picked up, and to additional pressure on Brazil assets in this morning's sell-off.

ETFs

- **Asset classes:** Equities were near-average ($23.4Bn, -0.2z), Fixed Income was strong ($15.2Bn, 0.6z), Commodities saw outflows (-$1.6Bn, -0.9z), and Currency/Multi-asset was roughly flat.
- **Regions:** US inflows were near average ($17Bn, -0.1z). International DM was below-average ($5.4Bn, -0.5z), including moderate outflows from Europe (-0.5z) but strong inflows to Canada (1.4z). Brazil flows were near-flat after two consecutive weeks of strong inflows into the final election round.
- **Sectors: Semis led for a second straight week**, with a pivot toward unlevered products. Semiconductor ETFs took in a record ~$12.7Bn (4.3z; SOXX + $6.5Bn, SOXL +$2.8Bn, SMH +$2.6Bn), surpassing the prior $9.3Bn record from late-July. Outside semis, flows were broadly weak again, led by **Utilities (-2.4z)**, Materials (-1.4z), Financials (-1.1z), and Staples (-1z).
- **Styles:** Defined outcome ($0.9Bn, 1.3z) drew strong inflows, while Multi-factor (-1.4z), Volatility (-0.8z) and Thematic (-0.7z) ETFs saw outflows.
- **Levered ETFs:** Levered products saw another strong week (~$4.3Bn, 1.7z), led by SOXL (+$2.8Bn, its 2nd highest ever, but well below last week's + $7.2Bn), TQQQ (+$1.3Bn), and SOXS (+$0.8Bn).
- **Fixed Income: Munis ($4.4Bn, 1.8z)** posted a 4th straight week of strong (though decelerating) inflows. **Medium-term gov't bonds ($2.8Bn, 2.1z)** and Corporates (0.7z) also drew strong demand, while Money Market (-$0.9Bn, -1.1z) and Loans (-1.2z) recorded outflows.
- **Commodities: Precious Metals (-$1.6Bn, -0.9z) saw strong outflows**, concentrated in GLD (-$2.3Bn). Agriculture also saw notable outflows (-1.3z).
- **Currency/Multi-Asset:** Liquid Alts posted their first weekly outflows in ~3 months (-1.5z), mainly from managed futures products.

See page 21 for analyst certification and important disclosures.

Global Equity Derivatives Strategy

Bram Kaplan, CFA AC
(1-212) 272-1215
bram.kaplan@jpmorgan.com
J.P. Morgan Securities LLC

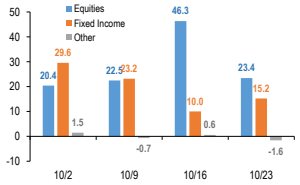
Daniel Motoc, CFA AC
(1-212) 622-0105
daniel.motoc@jpmchase.com
J.P. Morgan Securities LLC

Arda Sebuktekin
(1-212) 270-4397
arda.sebuktekin@jpmchase.com
J.P. Morgan Securities LLC

Tony SK Lee
(852) 2800-8857
tony.sk.lee@jpmorgan.com
J.P. Morgan Securities (Asia Pacific) Limited/ J.P. Morgan Broking (Hong Kong) Limited

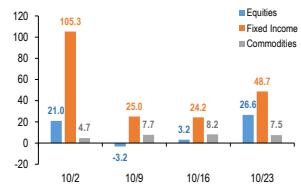
Davide Silvestrini
(44-20) 7134-4082
davide.silvestrini@jpmorgan.com
J.P. Morgan Securities plc

4w US ETF Flows ($Bn)



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

4w Global Futures Flows ($Bn)



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P. Sum of futures in Tables 1-5.

Bram Kaplan, CFA AC
(1-212) 272-1215
bram.kaplan@jpmorgan.com

J.P.Morgan

Futures

- **Flows:** Last week we saw large net buying (>1.5z) in S&P 500, FTSE 100, FTSE Taiwan, TAIEX, UST Long Bond, UK 10y and Copper, and large net selling (<-1.5z) in Dow Jones, SMI, China A50, NIFTY and CAD futures.
- **CTAs:** CTAs likely **sold US equities** as short-term signals turned negative on SPX/NDX and volatility rose. They also likely **sold Japan equities** and are near neutral across APAC overall (while still short China), and **sold Bovespa/BRL** on this morning's sell-off. In rates, they likely trimmed shorts as short-term momentum turned positive in the US and Europe. They also flipped short US Natgas, remain short metals (ex-Copper), and are long USD vs DM FX (ex-JPY).
- **CFTC Positioning:** Asset Managers added to **EAFE (1.5z)** and cut VIX shorts (2.1z) w/w, likely following the Oct 16 intraday VIX spike. They also extended duration, rotating out of UST 2y-10y into Ultra Bond futures. Leveraged funds significantly **cut RTY shorts (+3z)**, sold VIX (-1.9z), and **increased Ultra Bond shorts (-2.9z)**. Managed Money sold Copper (-1.9z), US Natgas, and Ags.

Bram Kaplan, CFA AC (1-212) 272-1215
bram.kaplan@jpmorgan.com
J.P. Morgan Securities LLC

Daniel Motoc, CFA AC (1-212) 622-0105
daniel.motoc@jpmchase.com

Arda Sebuktekin (1-212) 270-4397
arda.sebuktekin@jpmchase.com

Tony SK Lee (852) 2800-8857
tony.sk.lee@jpmorgan.com

Futures Flows & Positioning

Table 1: Equity Futures Flows & Positioning

See Appendix for definitions; asterisk denotes expiry week

Americas	Ticker	Net Flow - 1y History	1w Net Flow (z)	4w Net Flow (z)	1w Return	OI Change (w/w)	OI Correlation w/Spot	CTA Signal
S&P 500	ES1 Index		2.1	-0.2	-0.5%	1.2%	-9%	Long
Nasdaq 100	NQ1 Index		-0.5	0.7	-0.8%	2.3%	-13%	Neutral
Russell 2000	RTY1 Index		0.5	1.8	2.0%	0.2%	18%	Short
DJIA	DM1 Index		-1.6	-0.1	0.4%	0.3%	21%	Long
S&P MidCap 400	FA1 Index		-1.4	0.1	0.0%	-0.6%	-16%	Neutral
MSCI EM	MES1 Index		0.2	0.7	-2.6%	0.6%	-11%	Short
MSCI EAFE	MFS1 Index		-0.2	1.6	-1.5%	-0.8%	-8%	Neutral
VIX	UX1 Index		-0.3	0.7	-2.7%	-16.1% *	18%	Neutral
S&P TSX 60	PT1 Index		0.6	2.2	-0.6%	-0.4%	-11%	Long
Bovespa	XB1 Index		-0.3	-1.7	-0.2%	9.7%	-7%	Short
EMEA	**Ticker**	**Net Flow - 1y History**	**1w Net Flow (z)**	**4w Net Flow (z)**	**1w Return**	**OI Change (w/w)**	**OI Correlation w/Spot**	**CTA Signal**
Euro Stoxx 50	VG1 Index		0.7	0.2	-2.2%	2.8%	26%	Neutral
FTSE 100	Z 1 Index		1.9	1.0	-1.8%	-3.7%	44%	Long
DAX	GX1 Index		0.7	-0.4	-1.7%	6.3%	-5%	Long
SMI	SM1 Index		-1.8	-1.9	0.5%	4.0%	16%	Long
CAC 40	CF1 Index		0.2	-2.0	-2.2%	-4.2%	-20%	Short
FTSE/JSE Top 40	AI1 Index		0.2	-0.2	-3.1%	-2.2%	11%	Short
BIST 30	A51 Index		-0.7	-0.5	-4.7%	-9.4%	-4%	Short
APAC	**Ticker**	**Net Flow - 1y History**	**1w Net Flow (z)**	**4w Net Flow (z)**	**1w Return**	**OI Change (w/w)**	**OI Correlation w/Spot**	**CTA Signal**
TOPIX	TP1 Index		0.2	1.1	-2.5%	-0.8%	-16%	Neutral
Nikkei 225	NK1 Index		1.4	1.5	-4.0%	0.7%	-4%	Neutral
ASX SPI 200	XP1 Index		0.7	1.0	-0.1%	-0.9%	-8%	Neutral
KOSPI 200	KM1 Index		-1.0	-1.6	-1.8%	0.2%	0%	Short
Hang Seng	HI1 Index		-1.4	-1.7	-4.5%	6.1%	-30%	Short
Hang Seng China Ent.	HC1 Index		-0.3	-0.4	-4.6%	11.1%	-8%	Short
FTSE China A50	XU1 Index		-2.0	-1.7	-2.5%	5.8%	-14%	Short
CSI 300	IFB1 Index		-0.2	0.6	-1.6%	5.5%	-47%	Short
CSI 500	FFD1 Index		0.3	0.4	0.6%	6.7%	-36%	Short
CSI 1000	IFD1 Index		0.1	0.1	0.8%	5.7%	-31%	Short
SSE50	FFB1 Index		-0.3	0.2	-0.4%	1.9%	-1%	Short
NIFTY 50	JGS1 Index		-1.6	-1.0	-2.4%	4.7%	-34%	Short
SET 50	BC1 Index		0.0	-0.5	-0.1%	-2.3%	-1%	Long
SGX FTSE Taiwan	TWT1 Index		2.0	1.7	-2.6%	8.6%	8%	Neutral
TAIEX	FT1 Index		2.4	1.7	-2.3%	-2.7% *	-3%	Long
MSCI Singapore	QZ1 Index		-0.8	-0.5	-2.1%	0.0%	-1%	Neutral

Source: J.P. Morgan Equity Derivatives Strategy, CFTC, Bloomberg Finance L.P.

Bram Kaplan, CFA [AC] (1-212) 272-1215
bram.kaplan@jpmorgan.com
J.P. Morgan Securities LLC

Daniel Motoc, CFA [AC] (1-212) 622-0105
daniel.motoc@jpmchase.com

Arda Sebuktekin (1-212) 270-4397
arda.sebuktekin@jpmchase.com

Tony SK Lee (852) 2800-8857
tony.sk.lee@jpmorgan.com

Table 2: Fixed Income Futures Flows & Positioning

See Appendix for definitions; asterisk denotes expiry week

Americas	Ticker	Net Flow - 1y History	1w Net Flow (z)	1w Net Flow (DV01)	4w Net Flow (z)	1w Return	OI Change (w/w)	OI Correlation w/Spot	CTA Signal
2y UST	TU1 Comdty		0.1	-$30k/01	1.0	0.1%	-1.9%	-6%	Short
5y UST	FV1 Comdty		1.0	+$2,959k/01	3.1	0.4%	-1.1%	10%	Short
10y UST	TY1 Comdty		0.8	+$5,597k/01	2.6	0.7%	0.9%	16%	Short
UST Ultra 10y Note	UXY1 Comdty		-0.1	-$485k/01	3.3	0.8%	-2.4%	12%	Short
UST Long Bond	US1 Comdty		1.9	+$6,346k/01	0.8	1.7%	2.8%	16%	Short
UST Ultra Bond	WN1 Comdty		0.5	+$1,988k/01	1.6	1.8%	-3.8%	9%	Short
Canada 10y	CN1 Comdty		0.6	+$635k/01	-0.1	0.0%	1.2%	25%	Short
EMEA	**Ticker**	**Net Flow - 1y History**	**1w Net Flow (z)**	**1w Net Flow (DV01)**	**4w Net Flow (z)**	**1w Return**	**OI Change (w/w)**	**OI Correlation w/Spot**	**CTA Signal**
Germany 2y	DU1 Comdty		0.4	+$778k/01	0.4	0.1%	-1.0%	-15%	Short
Germany 5y	OE1 Comdty		0.7	+$2,999k/01	-0.5	0.2%	4.1%	-2%	Short
Germany 10y	RX1 Comdty		0.5	+$3,968k/01	-0.6	0.5%	1.8%	-24%	Short
Germany 30y	UB1 Comdty		-0.1	+$440k/01	-0.5	1.9%	0.6%	-14%	Neutral
10y Gilt	G 1 Comdty		1.7	+$3,714k/01	1.8	0.0%	0.2%	-7%	Short
France 10y	OAT1 Comdty		-0.6	-$459k/01	-0.1	-1.9%	10.3%	-23%	Short
Italy 10y	IK1 Comdty		0.5	+$1,057k/01	0.4	-0.8%	1.7%	-6%	Short
APAC	**Ticker**	**Net Flow - 1y History**	**1w Net Flow (z)**	**1w Net Flow (DV01)**	**4w Net Flow (z)**	**1w Return**	**OI Change (w/w)**	**OI Correlation w/Spot**	**CTA Signal**
Japan 10y	JB1 Comdty		0.3	+$3,324k/01	1.3	0.9%	-2.2%	-15%	Neutral
Australia 10y	XM1 Comdty		0.4	+$566k/01	-0.4	0.1%	0.9%	-1%	Short
China 10y	TFT1 Comdty		-0.8	-$1,171k/01	-0.2	0.1%	-0.7%	11%	Long
Korea 10y	KAA1 Comdty		1.1	+$501k/01	2.8	0.5%	-4.5%	-17%	Neutral

Source: J.P. Morgan Equity Derivatives Strategy, CFTC, Bloomberg Finance L.P.

Table 3: Interest Rates Futures Flows & Positioning

See Appendix for definitions; asterisk denotes expiry week

Global	Ticker	Net Flow - 1y History	1w Net Flow (z)	1w Net Flow (DV01)	4w Net Flow (z)	1w Chg (bps)	OI Change (w/w)	OI Correlation w/Spot	CTA Signal
3M SOFR	SFR1 Comdty		0.4	+$285k/01	1.6	0.50	0.6%	-15%	Short
3M Euribor	ER1 Comdty		1.4	+$556k/01	0.5	1.50	0.8%	-6%	Short
3M SONIA	SFI1 Comdty		0.5	+$174k/01	1.3	0.00	2.5%	13%	Short

Source: J.P. Morgan Equity Derivatives Strategy, CFTC, Bloomberg Finance L.P.

Bram Kaplan, CFA AC (1-212) 272-1215
bram.kaplan@jpmorgan.com
J.P. Morgan Securities LLC
Daniel Motoc, CFA AC (1-212) 622-0105
daniel.motoc@jpmchase.com
Arda Sebuktekin (1-212) 270-4397
arda.sebuktekin@jpmchase.com
Tony SK Lee (852) 2800-8857
tony.sk.lee@jpmorgan.com

Table 4: Commodity Futures Flows & Positioning

See Appendix for definitions; asterisk denotes expiry week

Energy	Ticker	Net Flow - 1y History	1w Net Flow (z)	4w Net Flow (z)	1w Return	OI Change (w/w)	OI Correlation w/Spot	CTA Signal
Brent Crude	CO1 Comdty		1.4	1.4	6.2%	3.2%	19%	Long
WTI	CL1 Comdty		0.3	1.5	1.8%	2.4% *	35%	Long
US Natural Gas	NG1 Comdty		0.6	0.9	-15.3%	-1.3%	-5%	Short
Gasoline	XB1 Comdty		0.6	-0.2	-0.7%	-0.2%	-1%	Neutral
Heating Oil	HO1 Comdty		-0.1	0.0	1.7%	-3.9%	19%	Long
Low Sulphur Gasoil	QS1 Comdty		-0.3	1.0	5.6%	-1.3%	-5%	Long
TTF Nat Gas	TZT1 Comdty		1.2	2.2	7.2%	1.8%	-5%	Long
Base Metals	**Ticker**	**Net Flow - 1y History**	**1w Net Flow (z)**	**4w Net Flow (z)**	**1w Return**	**OI Change (w/w)**	**OI Correlation w/Spot**	**CTA Signal**
Copper	HG1 Comdty		2.8	3.1	-1.0%	4.5%	-11%	Long
Nickel	LN1 Comdty		-0.8	-0.6	3.6%	-7.1% *	6%	Short
Aluminum	LA1 Comdty		-0.3	0.5	2.5%	-3.1% *	7%	Short
Precious Metals	**Ticker**	**Net Flow - 1y History**	**1w Net Flow (z)**	**4w Net Flow (z)**	**1w Return**	**OI Change (w/w)**	**OI Correlation w/Spot**	**CTA Signal**
Gold	GC1 Comdty		0.2	0.2	-4.0%	0.4%	5%	Short
Silver	SI1 Comdty		0.4	0.5	-5.9%	2.6%	10%	Short
Agriculture	**Ticker**	**Net Flow - 1y History**	**1w Net Flow (z)**	**4w Net Flow (z)**	**1w Return**	**OI Change (w/w)**	**OI Correlation w/Spot**	**CTA Signal**
Soybean	S 1 Comdty		-0.1	-1.0	-0.4%	-5.1%	9%	Long
Corn	C 1 Comdty		-1.5	-0.9	-3.8%	-0.6%	5%	Short
Wheat	W 1 Comdty		0.3	0.2	-5.4%	2.3%	-34%	Short

Source: J.P. Morgan Equity Derivatives Strategy, CFTC, Bloomberg Finance L.P.

Table 5: FX/Crypto Futures Flows & Positioning

See Appendix for definitions; asterisk denotes expiry week

FX/Crypto	Ticker	Net Flow - 1y History	1w Net Flow (z)	4w Net Flow (z)	1w Return	OI Change (w/w)	OI Correlation w/Spot	CTA Signal
EUR/USD	EC1 Curncy		-0.3	-0.1	-0.2%	-0.6%	1%	Short
GBP/USD	BP1 Curncy		-0.9	-0.7	0.5%	-2.3%	0%	Short
JPY/USD	JY1 Curncy		-0.8	0.2	1.0%	9.4%	11%	Long
CAD/USD	CD1 Curncy		-2.3	-0.1	0.7%	-4.7%	11%	Short
AUD/USD	AD1 Curncy		-0.1	1.1	0.8%	0.3%	11%	Neutral
USD/CNH	XUC1 Curncy		-0.2	-0.4	-0.3%	4.2%	19%	Short
CHF/USD	SF1 Curncy		-0.6	-1.2	0.1%	0.6%	0%	Short
USD/BRL	UC1 Curncy		-0.4	1.3	-1.1%	6.1%	7%	Long
Bitcoin	BTC1 Curncy		-0.2	-0.5	-0.8%	3.4%	27%	Neutral
Ether	DCR1 Curncy		-0.3	-1.1	-0.1%	11.3%	29%	Neutral

Source: J.P. Morgan Equity Derivatives Strategy, CFTC, Bloomberg Finance L.P.

Bram Kaplan, CFA AC (1-212) 272-1215
bram.kaplan@jpmorgan.com
J.P. Morgan Securities LLC

Daniel Motoc, CFA AC (1-212) 622-0105
daniel.motoc@jpmchase.com

Arda Sebuktekin (1-212) 270-4397
arda.sebuktekin@jpmchase.com

Tony SK Lee (852) 2800-8857
tony.sk.lee@jpmorgan.com

CTA Signals

Table 6: Equity CTA Signals

"Current" levels are as of morning ET Oct 26; see Appendix for definitions

				Current CTA Levels								CTA Levels in 1w								
Americas	**Ticker**	**Current**	**Prior Week CTA Signal**	**1m Mom.**	**3m Mom.**	**6m Mom.**	**12m Mom.**	**50d MA**	**100d MA**	**200d MA**	**CTA Signal**	**1m Mom.**	**3m Mom.**	**6m Mom.**	**12m Mom.**	**50d MA**	**100d MA**	**200d MA**	**CTA Signal**	**1w Chg**
S&P 500	ES1 Index	7,691	Long	7,781 -1.2%	7,454 3.2%	7,191 7.0%	6,837 12.5%	7,732 -0.5%	7,637 0.7%	7,319 5.1%	Long	7,739 -0.6%	7,638 0.7%	7,259 6.0%	6,871 11.9%	7,727 -0.5%	7,652 0.5%	7,340 4.8%	Long	-
Nasdaq 100	NQ1 Index	29,238	Long	30,807 -5.1%	28,131 3.9%	27,348 6.9%	25,576 14.3%	29,975 -2.5%	29,719 -1.6%	27,959 4.6%	Neutral	30,840 -5.2%	29,053 0.6%	27,916 4.7%	25,997 12.5%	29,923 -2.3%	29,700 -1.6%	28,062 4.2%	Neutral	-
Russell 2000	RTY1 Index	2,794	Short	2,859 -2.3%	2,955 -5.4%	2,788 0.2%	2,516 11.0%	2,892 -3.4%	2,934 -4.8%	2,810 -0.5%	Short	2,832 -1.3%	2,991 -6.6%	2,826 -1.1%	2,482 12.6%	2,863 -2.4%	2,930 -4.6%	2,814 -0.7%	Short	↓
DJIA	DM1 Index	52,587	Neutral	51,918 1.3%	52,483 0.2%	49,344 6.6%	47,343 11.1%	52,491 0.2%	52,488 0.2%	50,677 3.8%	Long	51,332 2.4%	53,412 -1.5%	49,380 6.5%	47,624 10.4%	52,391 0.4%	52,591 0.0%	50,767 3.6%	Long	↓
S&P MidCap 400	FA1 Index	3,657	Short	3,673 -0.4%	3,808 -4.0%	3,642 0.4%	3,311 10.4%	3,720 -1.7%	3,764 -2.9%	3,657 0.0%	Neutral	3,657 0.0%	3,824 -4.4%	3,651 0.2%	3,250 12.5%	3,694 -1.0%	3,760 -2.7%	3,661 -0.1%	Neutral	-
MSCI EM	MES1 Index	1,567	Short	1,735 -9.7%	1,617 -3.1%	1,624 -3.5%	1,406 11.4%	1,702 -7.9%	1,696 -7.6%	1,637 -4.3%	Short	1,718 -8.8%	1,660 -5.6%	1,650 -5.0%	1,412 11.0%	1,686 -7.1%	1,689 -7.2%	1,639 -4.4%	Short	-
MSCI EAFE	MFS1 Index	3,101	Neutral	3,165 -2.0%	3,131 -0.9%	3,019 2.7%	2,831 9.5%	3,176 -2.3%	3,164 -2.0%	3,100 0.0%	Neutral	3,124 -0.7%	3,204 -3.2%	3,024 2.6%	2,810 10.4%	3,158 -1.8%	3,166 -2.0%	3,103 -0.1%	Short	-
VIX	UX1 Index	19.10	Neutral	17.68 8.0%	19.12 -0.1%	20.21 -5.5%	18.53 3.1%	17.30 10.4%	17.65 8.2%	19.09 0.0%	Neutral	17.87 6.9%	18.03 5.9%	19.81 -3.6%	19.06 0.2%	17.64 8.3%	17.66 8.2%	19.17 -0.4%	Long	-
S&P TSX 60	PT1 Index	2,141	Long	2,101 1.9%	2,103 1.8%	1,968 8.8%	1,785 19.9%	2,120 1.0%	2,098 2.0%	2,018 6.1%	Long	2,079 3.0%	2,094 2.2%	1,968 8.8%	1,781 20.2%	2,119 1.0%	2,105 1.7%	2,025 5.7%	Long	-
Bovespa	XB1 Index	172,585	Long	185,562 -7.0%	176,405 -2.2%	193,001 -10.6%	149,122 15.7%	189,739 -9.0%	182,015 -5.2%	183,036 -5.7%	Short	189,630 -9.0%	178,585 -3.4%	189,471 -8.9%	152,104 13.5%	190,022 -9.2%	182,165 -5.3%	183,260 -5.8%	Short	-
EMEA	**Ticker**	**Current**	**Prior Week CTA Signal**	**1m Mom.**	**3m Mom.**	**6m Mom.**	**12m Mom.**	**50d MA**	**100d MA**	**200d MA**	**CTA Signal**	**1m Mom.**	**3m Mom.**	**6m Mom.**	**12m Mom.**	**50d MA**	**100d MA**	**200d MA**	**CTA Signal**	**1w Chg**
Euro Stoxx 50	VG1 Index	6,123	Neutral	6,315 -3.0%	6,312 -3.0%	5,814 5.3%	5,692 7.6%	6,315 -3.0%	6,322 -3.2%	6,108 0.2%	Neutral	6,253 -2.1%	6,446 -5.0%	5,802 5.5%	5,686 7.7%	6,272 -2.4%	6,323 -3.2%	6,112 0.2%	Neutral	-
FTSE 100	Z 1 Index	10,791	Long	10,755 0.3%	10,806 -0.1%	10,361 4.2%	9,658 11.7%	10,741 0.5%	10,681 1.0%	10,526 2.5%	Long	10,550 2.3%	10,863 -0.7%	10,326 4.5%	9,752 10.7%	10,740 0.5%	10,708 0.8%	10,545 2.3%	Long	-
DAX	GX1 Index	25,545	Neutral	25,537 0.0%	25,427 0.5%	24,217 5.5%	24,347 4.9%	25,721 -0.7%	25,546 0.0%	24,972 2.3%	Long	25,293 1.0%	26,051 -1.9%	24,333 5.0%	24,147 5.8%	25,641 -0.4%	25,607 -0.2%	24,984 2.2%	Neutral	↓
SMI	SM1 Index	14,503	Long	13,930 4.1%	14,449 0.4%	13,118 10.6%	12,554 15.5%	14,109 2.8%	14,145 2.5%	13,663 6.1%	Long	13,716 5.7%	14,402 0.7%	13,029 11.3%	12,264 18.3%	14,126 2.7%	14,203 2.1%	13,699 5.9%	Long	-
CAC 40	CF1 Index	7,871	Short	8,091 -2.7%	8,428 -6.6%	8,056 -2.3%	8,245 -4.5%	8,122 -3.1%	8,283 -5.0%	8,211 -4.1%	Short	7,902 -0.4%	8,609 -8.6%	8,004 -1.7%	8,141 -3.3%	8,046 -2.2%	8,260 -4.7%	8,202 -4.0%	Short	-
FTSE/JSE Top 40	AI1 Index	101,713	Short	104,440 -2.6%	101,982 -0.3%	109,013 -6.7%	103,584 -1.8%	105,291 -3.4%	104,501 -2.7%	107,730 -5.6%	Short	101,438 0.3%	104,556 -2.7%	107,759 -5.6%	102,845 -1.1%	104,615 -2.8%	104,372 -2.5%	107,394 -5.3%	Short	↑
BIST 30	A51 Index	13,879	Short	16,427 -15.5%	16,086 -13.7%	16,658 -16.7%	11,857 17.0%	16,165 -14.1%	16,294 -14.8%	15,962 -13.0%	Short	15,441 -10.1%	15,644 -11.3%	17,043 -18.6%	12,077 14.9%	15,843 -12.4%	16,180 -14.2%	15,963 -13.1%	Short	-
APAC	**Ticker**	**Current**	**Prior Week CTA Signal**	**1m Mom.**	**3m Mom.**	**6m Mom.**	**12m Mom.**	**50d MA**	**100d MA**	**200d MA**	**CTA Signal**	**1m Mom.**	**3m Mom.**	**6m Mom.**	**12m Mom.**	**50d MA**	**100d MA**	**200d MA**	**CTA Signal**	**1w Chg**
TOPIX	TP1 Index	3,981	Long	4,085 -2.6%	4,023 -1.1%	3,742 6.4%	3,286 21.1%	4,095 -2.8%	4,047 -1.7%	3,874 2.7%	Neutral	4,119 -3.4%	3,966 0.4%	3,721 7.0%	3,326 19.7%	4,073 -2.3%	4,049 -1.7%	3,890 2.3%	Neutral	↑
Nikkei 225	NK1 Index	62,200	Long	65,620 -5.2%	64,043 -2.9%	59,993 3.7%	49,527 25.6%	66,269 -6.1%	66,770 -6.8%	61,527 1.1%	Neutral	68,200 -8.8%	63,817 -2.5%	59,420 4.7%	52,133 19.3%	65,561 -5.1%	66,470 -6.4%	61,860 0.5%	Neutral	-
ASX SPI 200	XP1 Index	8,827	Neutral	8,755 0.8%	8,822 0.1%	8,771 0.6%	9,054 -2.5%	8,852 -0.3%	8,856 -0.3%	8,820 0.1%	Neutral	8,722 1.2%	9,001 -1.9%	8,715 1.3%	8,898 -0.8%	8,830 0.0%	8,864 -0.4%	8,821 0.1%	Neutral	-
KOSPI 200	KM1 Index	850.0	Short	1,127.8 -24.6%	1,031.6 -17.6%	990.9 -14.2%	551.7 54.1%	1,039.6 -18.2%	1,130.2 -24.8%	999.5 -15.0%	Short	1,104.2 -23.0%	1,007.7 -15.6%	1,033.2 -17.7%	584.8 45.4%	1,016.5 -16.4%	1,099.7 -22.7%	1,006.3 -15.5%	Short	-
Hang Seng	HI1 Index	23,236	Short	24,676 -5.8%	25,173 -7.7%	25,867 -10.2%	26,185 -11.3%	24,747 -6.1%	24,642 -5.7%	25,318 -8.2%	Short	24,355 -4.6%	25,923 -10.4%	25,814 -10.0%	26,140 -11.1%	24,465 -5.0%	24,564 -5.4%	25,216 -7.9%	Short	-
Hang Seng China Ent.	HC1 Index	7,808	Short	8,226 -5.1%	8,364 -6.6%	8,727 -10.5%	9,379 -16.7%	8,255 -5.4%	8,214 -4.9%	8,497 -8.1%	Short	8,135 -4.0%	8,622 -9.4%	8,717 -10.4%	9,274 -15.8%	8,173 -4.5%	8,183 -4.6%	8,457 -7.7%	Short	-
FTSE China A50	XU1 Index	13,106	Short	14,378 -8.8%	15,022 -12.8%	15,557 -15.8%	15,490 -15.4%	14,215 -7.8%	14,682 -10.7%	14,898 -12.0%	Short	13,878 -5.6%	14,726 -11.0%	15,652 -16.3%	15,376 -14.8%	14,005 -6.4%	14,545 -9.9%	14,841 -11.7%	Short	-
CSI 300	IFB1 Index	4,067	Short	4,459 -8.8%	4,605 -11.7%	4,750 -14.4%	4,640 -12.3%	4,460 -8.8%	4,609 -11.8%	4,652 -12.6%	Short	4,353 -6.6%	4,564 -10.9%	4,799 -15.3%	4,666 -12.8%	4,388 -7.3%	4,561 -10.8%	4,635 -12.2%	Short	-
CSI 500	FFD1 Index	6,678	Short	7,653 -12.7%	7,513 -11.1%	8,227 -18.8%	7,186 -7.1%	7,554 -11.6%	7,857 -15.0%	8,011 -16.6%	Short	7,429 -10.1%	7,491 -10.9%	8,341 -19.9%	7,320 -8.8%	7,400 -9.8%	7,760 -13.9%	7,990 -16.4%	Short	-
CSI 1000	IFD1 Index	6,601	Short	7,602 -13.2%	7,038 -6.2%	8,266 -20.1%	7,319 -9.8%	7,412 -10.9%	7,703 -14.3%	7,932 -16.8%	Short	7,299 -9.6%	7,141 -7.6%	8,362 -21.1%	7,460 -11.5%	7,279 -9.3%	7,597 -13.1%	7,905 -16.5%	Short	-
SSE50	FFB1 Index	2,737	Short	2,854 -4.1%	2,917 -6.2%	2,941 -6.9%	3,043 -10.1%	2,858 -4.2%	2,882 -5.0%	2,936 -6.8%	Short	2,817 -2.8%	2,887 -5.2%	2,980 -8.2%	3,028 -9.6%	2,834 -3.4%	2,873 -4.7%	2,927 -6.5%	Short	-
NIFTY 50	JGS1 Index	22,755	Short	23,215 -2.0%	23,917 -4.9%	24,080 -5.5%	25,980 -12.4%	23,400 -2.8%	23,744 -4.2%	24,049 -5.4%	Short	22,635 0.5%	24,581 -7.4%	24,169 -5.9%	25,914 -12.2%	23,216 -2.0%	23,713 -4.0%	23,966 -5.1%	Short	↑
SET 50	BC1 Index	1,067	Neutral	1,079 -1.1%	1,087 -1.8%	962 10.9%	856 24.7%	1,066 0.1%	1,059 0.8%	997 7.0%	Long	1,044 2.3%	1,082 -1.4%	973 9.7%	850 25.6%	1,065 0.2%	1,061 0.6%	1,004 6.3%	Long	↑
SGX FTSE Taiwan	TWT1 Index	3,983	Long	4,260 -6.5%	3,684 8.1%	3,257 22.3%	2,272 75.3%	4,124 -3.4%	4,002 -0.5%	3,502 13.7%	Neutral	4,282 -7.0%	3,730 6.8%	3,419 16.5%	2,312 72.3%	4,134 -3.7%	4,011 -0.7%	3,544 12.4%	Neutral	-
TAIEX	FT1 Index	46,234	Long	48,187 -4.1%	43,089 7.3%	39,625 16.7%	27,777 66.4%	46,921 -1.5%	45,938 0.6%	40,457 14.3%	Long	48,566 -4.8%	43,430 6.5%	40,477 14.2%	28,390 62.9%	46,979 -1.6%	45,943 0.6%	40,895 13.1%	Long	-
MSCI Singapore	QZ1 Index	493	Neutral	532.2 -7.4%	519.9 -5.3%	442.1 11.4%	445.7 10.5%	526.6 -6.5%	513.8 -4.1%	482.4 2.1%	Neutral	528.8 -6.9%	519.8 -5.2%	444.2 10.9%	446.8 10.2%	521.3 -5.5%	516.0 -4.5%	483.6 1.9%	Neutral	-

Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

Bram Kaplan, CFA [AC] (1-212) 272-1215
bram.kaplan@jpmorgan.com
J.P. Morgan Securities LLC

Daniel Motoc, CFA [AC] (1-212) 622-0105
daniel.motoc@jpmchase.com

Arda Sebuktekin (1-212) 270-4397
arda.sebuktekin@jpmchase.com

Tony SK Lee (852) 2800-8857
tony.sk.lee@jpmorgan.com

Table 7: Fixed Income CTA Signals

"Current" levels are as of morning ET Oct 26; see Appendix for definitions

				Current CTA Levels								CTA Levels in 1w								
Americas	**Ticker**	**Current**	**Prior Week CTA Signal**	**1m Mom.**	**3m Mom.**	**6m Mom.**	**12m Mom.**	**50d MA**	**100d MA**	**200d MA**	**CTA Signal**	**1m Mom.**	**3m Mom.**	**6m Mom.**	**12m Mom.**	**50d MA**	**100d MA**	**200d MA**	**CTA Signal**	**1w Chg**
2y UST	TU1 Comdty	102.1	Short	101.9 0.1%	102.9 -0.8%	103.6 -1.4%	104.3 -2.1%	102.3 -0.2%	102.7 -0.6%	103.3 -1.1%	Short	101.8 0.3%	103.0 -0.9%	103.5 -1.4%	104.1 -2.0%	102.2 -0.1%	102.6 -0.5%	103.2 -1.1%	Short	-
5y UST	FV1 Comdty	104.5	Short	103.7 0.8%	106.4 -1.8%	107.8 -3.1%	109.6 -4.6%	104.8 -0.2%	105.7 -1.1%	107.1 -2.4%	Short	103.5 1.0%	106.4 -1.8%	107.8 -3.0%	109.2 -4.3%	104.5 0.0%	105.6 -1.0%	106.9 -2.3%	Short	-
10y UST	TY1 Comdty	106.2	Short	104.6 1.5%	108.7 -2.3%	110.6 -4.0%	113.3 -6.2%	106.3 -0.1%	107.8 -1.4%	109.6 -3.1%	Short	104.4 1.8%	108.7 -2.3%	110.6 -3.9%	112.6 -5.7%	106.1 0.1%	107.6 -1.3%	109.4 -2.9%	Short	↑
UST Ultra 10y Note	UXY1 Comdty	107.7	Short	105.6 2.1%	110.7 -2.7%	112.9 -4.6%	116.3 -7.4%	107.7 0.0%	109.6 -1.7%	111.7 -3.6%	Short	105.1 2.5%	110.6 -2.6%	112.9 -4.6%	115.3 -6.6%	107.4 0.3%	109.3 -1.5%	111.5 -3.4%	Short	-
UST Long Bond	US1 Comdty	107.6	Short	103.9 3.6%	110.2 -2.4%	113.0 -4.8%	118.7 -9.4%	106.6 0.9%	109.0 -1.3%	111.8 -3.8%	Short	102.5 4.9%	109.7 -2.0%	113.2 -5.0%	117.0 -8.1%	106.5 1.0%	108.7 -1.1%	111.5 -3.6%	Short	-
UST Ultra Bond	WN1 Comdty	109.3	Short	105.0 4.1%	111.9 -2.3%	115.3 -5.2%	123.2 -11.3%	108.0 1.3%	110.7 -1.2%	113.9 -4.0%	Short	103.4 5.8%	111.1 -1.6%	115.4 -5.3%	120.8 -9.5%	107.9 1.4%	110.4 -1.0%	113.6 -3.7%	Short	-
Canada 10y	CN1 Comdty	116.8	Short	115.4 1.2%	119.6 -2.3%	119.1 -1.9%	123.1 -5.1%	117.0 -0.2%	118.5 -1.4%	119.6 -2.3%	Short	115.8 0.9%	119.4 -2.2%	119.2 -2.0%	122.6 -4.7%	116.8 0.0%	118.3 -1.2%	119.4 -2.2%	Short	-
EMEA	**Ticker**	**Current**	**Prior Week CTA Signal**	**1m Mom.**	**3m Mom.**	**6m Mom.**	**12m Mom.**	**50d MA**	**100d MA**	**200d MA**	**CTA Signal**	**1m Mom.**	**3m Mom.**	**6m Mom.**	**12m Mom.**	**50d MA**	**100d MA**	**200d MA**	**CTA Signal**	**1w Chg**
Germany 2y	DU1 Comdty	105.0	Short	104.5 0.5%	105.5 -0.5%	105.7 -0.7%	107.1 -2.0%	105.0 0.0%	105.3 -0.3%	105.8 -0.8%	Short	104.9 0.0%	105.6 -0.6%	105.8 -0.8%	107.0 -1.9%	104.9 0.1%	105.3 -0.3%	105.7 -0.7%	Short	-
Germany 5y	OE1 Comdty	113.1	Short	111.7 1.2%	114.0 -0.9%	115.3 -1.9%	118.3 -4.4%	112.8 0.2%	113.7 -0.5%	114.8 -1.6%	Short	112.7 0.3%	114.3 -1.1%	115.6 -2.2%	118.1 -4.3%	112.7 0.3%	113.6 -0.4%	114.7 -1.5%	Short	-
Germany 10y	RX1 Comdty	122.1	Short	119.5 2.2%	124.8 -2.2%	125.1 -2.4%	129.6 -5.8%	121.7 0.3%	123.6 -1.3%	125.2 -2.5%	Short	120.9 1.0%	125.1 -2.4%	125.5 -2.7%	129.2 -5.5%	121.5 0.5%	123.4 -1.1%	125.0 -2.3%	Short	-
Germany 30y	UB1 Comdty	105.6	Short	100.3 5.3%	106.9 -1.2%	108.4 -2.6%	116.8 -9.6%	102.8 2.7%	105.4 0.2%	107.5 -1.8%	Neutral	102.1 3.5%	107.0 -1.3%	108.6 -2.8%	115.6 -8.6%	103.0 2.6%	105.3 0.4%	107.4 -1.6%	Neutral	-
10y Gilt	G 1 Comdty	85.3	Short	84.2 1.2%	87.0 -1.9%	86.5 -1.5%	93.7 -9.0%	85.1 0.2%	86.5 -1.4%	87.9 -3.0%	Short	84.3 1.2%	87.7 -2.8%	87.0 -2.0%	93.4 -8.7%	85.0 0.4%	86.3 -1.2%	87.7 -2.8%	Short	-
France 10y	OAT1 Comdty	108.6	Short	109.8 -1.1%	117.5 -7.6%	119.0 -8.7%	122.7 -11.5%	112.0 -3.0%	115.2 -5.7%	117.8 -7.8%	Short	109.2 -0.5%	117.9 -7.9%	119.6 -9.2%	122.4 -11.3%	111.1 -2.2%	114.6 -5.2%	117.5 -7.5%	Short	-
Italy 10y	IK1 Comdty	111.5	Short	110.8 0.7%	116.6 -4.3%	116.7 -4.4%	121.4 -8.1%	112.7 -1.1%	115.2 -3.2%	117.1 -4.7%	Short	110.6 0.8%	117.3 -4.9%	117.7 -5.2%	121.3 -8.0%	112.2 -0.6%	114.8 -2.9%	116.8 -4.5%	Short	-
APAC	**Ticker**	**Current**	**Prior Week CTA Signal**	**1m Mom.**	**3m Mom.**	**6m Mom.**	**12m Mom.**	**50d MA**	**100d MA**	**200d MA**	**CTA Signal**	**1m Mom.**	**3m Mom.**	**6m Mom.**	**12m Mom.**	**50d MA**	**100d MA**	**200d MA**	**CTA Signal**	**1w Chg**
Japan 10y	JB1 Comdty	126.7	Short	124.3 1.9%	127.2 -0.4%	129.6 -2.2%	136.1 -6.9%	125.6 0.9%	126.7 0.0%	128.8 -1.6%	Neutral	124.4 1.8%	127.0 -0.3%	129.6 -2.2%	136.0 -6.9%	125.6 0.8%	126.5 0.1%	128.6 -1.5%	Neutral	-
Australia 10y	XM1 Comdty	94.8	Short	94.6 0.2%	95.0 -0.2%	94.9 -0.1%	95.8 -1.0%	94.7 0.1%	94.9 -0.1%	95.0 -0.2%	Short	94.6 0.2%	95.0 -0.2%	95.0 -0.2%	95.7 -0.9%	94.7 0.1%	94.9 -0.1%	95.0 -0.2%	Short	-
China 10y	TFT1 Comdty	109.9	Long	109.6 0.3%	109.2 0.6%	108.6 1.2%	108.4 1.4%	109.5 0.3%	109.4 0.5%	108.9 0.9%	Long	109.4 0.4%	109.4 0.4%	108.6 1.2%	108.6 1.1%	109.6 0.3%	109.4 0.4%	108.9 0.9%	Long	-
Korea 10y	KAA1 Comdty	107.5	Neutral	104.0 3.4%	105.6 1.8%	108.9 -1.3%	117.3 -8.3%	105.2 2.2%	105.6 1.8%	108.0 -0.4%	Neutral	104.8 2.6%	106.5 1.0%	108.6 -1.0%	115.9 -7.2%	105.4 2.0%	105.7 1.7%	107.8 -0.3%	Neutral	-

Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

Table 8: Interest Rates CTA Signals

"Current" levels are as of morning ET Oct 26; see Appendix for definitions

				Current CTA Levels								CTA Levels in 1w								
Global	**Ticker**	**Current**	**Prior Week CTA Signal**	**1m Mom.**	**3m Mom.**	**6m Mom.**	**12m Mom.**	**50d MA**	**100d MA**	**200d MA**	**CTA Signal**	**1m Mom.**	**3m Mom.**	**6m Mom.**	**12m Mom.**	**50d MA**	**100d MA**	**200d MA**	**CTA Signal**	**1w Chg**
3M SOFR	SFR1 Comdty	96.1	Short	96.0 0.1%	96.3 -0.3%	96.3 -0.3%	95.9 0.1%	96.2 -0.1%	96.3 -0.2%	96.3 -0.2%	Short	96.1 0.0%	96.4 -0.3%	96.3 -0.3%	95.9 0.2%	96.1 -0.1%	96.2 -0.2%	96.3 -0.2%	Short	-
Dec 26 expiry	SFR2 Comdty	95.8	Short	95.6 0.2%	96.1 -0.3%	96.3 -0.5%	96.3 -0.5%	95.9 -0.1%	96.0 -0.2%	96.2 -0.4%	Short	95.8 0.0%	96.1 -0.4%	96.4 -0.6%	96.2 -0.4%	95.9 -0.1%	96.0 -0.2%	96.2 -0.4%	Short	-
Mar 27 expiry	SFR3 Comdty	95.6	Short	95.3 0.2%	95.9 -0.4%	96.3 -0.8%	96.5 -1.0%	95.7 -0.1%	95.8 -0.3%	96.1 -0.6%	Short	95.5 0.1%	96.0 -0.4%	96.3 -0.8%	96.4 -0.9%	95.6 0.0%	95.8 -0.2%	96.1 -0.5%	Short	-
Jun 27 expiry	SFR4 Comdty	95.4	Short	95.2 0.2%	95.9 -0.5%	96.3 -0.9%	96.8 -1.4%	95.5 -0.1%	95.7 -0.3%	96.1 -0.7%	Short	95.3 0.1%	95.9 -0.5%	96.3 -0.9%	96.6 -1.3%	95.5 0.0%	95.7 -0.3%	96.1 -0.7%	Short	-
Sep 27 expiry	SFR5 Comdty	95.4	Short	95.1 0.3%	95.8 -0.5%	96.3 -0.9%	96.9 -1.6%	95.5 -0.1%	95.7 -0.3%	96.1 -0.8%	Short	95.2 0.2%	95.9 -0.5%	96.2 -0.9%	96.8 -1.5%	95.4 0.0%	95.7 -0.3%	96.1 -0.7%	Short	-
Dec 27 expiry	SFR6 Comdty	95.4	Short	95.1 0.3%	95.9 -0.5%	96.3 -0.9%	97.0 -1.6%	95.5 -0.1%	95.7 -0.3%	96.1 -0.8%	Short	95.2 0.2%	95.9 -0.5%	96.2 -0.9%	96.9 -1.5%	95.4 0.0%	95.7 -0.3%	96.1 -0.7%	Short	-
Mar 28 expiry	SFR7 Comdty	95.4	Short	95.2 0.3%	95.9 -0.5%	96.3 -1.0%	97.0 -1.6%	95.5 -0.1%	95.7 -0.3%	96.1 -0.8%	Short	95.2 0.2%	96.0 -0.6%	96.3 -0.9%	96.9 -1.5%	95.4 0.0%	95.7 -0.3%	96.1 -0.7%	Short	-
Jun 28 expiry	SFR8 Comdty	95.4	Short	95.2 0.3%	96.0 -0.6%	96.4 -1.0%	97.0 -1.6%	95.5 -0.1%	95.8 -0.3%	96.2 -0.8%	Short	95.2 0.3%	96.0 -0.6%	96.4 -0.9%	96.9 -1.5%	95.4 0.0%	95.7 -0.3%	96.1 -0.7%	Short	-
Sep 28 expiry	SFR9 Comdty	95.5	Short	95.2 0.3%	96.0 -0.5%	96.4 -1.0%	96.9 -1.5%	95.5 0.0%	95.8 -0.3%	96.2 -0.7%	Short	95.2 0.3%	96.0 -0.6%	96.4 -1.0%	96.8 -1.4%	95.5 0.0%	95.8 -0.3%	96.1 -0.7%	Short	-
Dec 28 expiry	SFR10 Comdty	95.5	Short	95.2 0.3%	96.0 -0.5%	96.4 -1.0%	96.9 -1.4%	95.5 0.0%	95.8 -0.3%	96.2 -0.7%	Short	95.2 0.3%	96.0 -0.5%	96.4 -0.9%	96.8 -1.3%	95.5 0.0%	95.8 -0.3%	96.1 -0.7%	Short	-
Mar 29 expiry	SFR11 Comdty	95.5	Short	95.2 0.3%	96.0 -0.5%	96.4 -0.9%	96.8 -1.4%	95.5 0.0%	95.8 -0.3%	96.2 -0.7%	Short	95.2 0.3%	96.0 -0.5%	96.4 -0.9%	96.7 -1.3%	95.5 0.0%	95.8 -0.3%	96.1 -0.6%	Short	-
Jun 29 expiry	SFR12 Comdty	95.5	Short	95.2 0.3%	96.0 -0.5%	96.4 -0.9%	96.8 -1.3%	95.5 0.0%	95.8 -0.3%	96.1 -0.6%	Short	95.2 0.3%	96.0 -0.5%	96.4 -0.9%	96.7 -1.2%	95.5 0.0%	95.8 -0.3%	96.1 -0.6%	Short	-
Sep 29 expiry	SFR13 Comdty	95.5	Short	95.3 0.3%	96.0 -0.5%	96.4 -0.8%	96.7 -1.2%	95.5 0.0%	95.8 -0.3%	96.1 -0.6%	Short	95.2 0.3%	96.0 -0.5%	96.3 -0.8%	96.6 -1.1%	95.5 0.0%	95.8 -0.2%	96.1 -0.6%	Short	-
Dec 29 expiry	SFR14 Comdty	95.5	Short	95.3 0.3%	96.0 -0.5%	96.3 -0.8%	96.7 -1.2%	95.5 0.0%	95.8 -0.3%	96.1 -0.6%	Short	95.2 0.3%	96.0 -0.4%	96.3 -0.8%	96.6 -1.1%	95.5 0.1%	95.8 -0.2%	96.1 -0.6%	Short	-
Mar 30 expiry	SFR15 Comdty	95.5	Short	95.3 0.3%	96.0 -0.4%	96.3 -0.8%	96.7 -1.1%	95.5 0.0%	95.8 -0.3%	96.1 -0.6%	Short	95.2 0.3%	96.0 -0.4%	96.3 -0.8%	96.6 -1.1%	95.5 0.1%	95.8 -0.2%	96.1 -0.5%	Short	-
Jun 30 expiry	SFR16 Comdty	95.5	Short	95.3 0.3%	96.0 -0.4%	96.3 -0.8%	96.6 -1.1%	95.5 0.0%	95.8 -0.3%	96.1 -0.5%	Short	95.2 0.3%	95.9 -0.4%	96.2 -0.7%	96.5 -1.0%	95.5 0.1%	95.7 -0.2%	96.0 -0.5%	Short	-
3M Euribor	ER1 Comdty	97.1	Short	97.0 0.1%	97.4 -0.3%	97.6 -0.5%	98.0 -0.9%	97.2 -0.1%	97.3 -0.2%	97.5 -0.4%	Short	97.1 0.0%	97.4 -0.3%	97.6 -0.5%	98.0 -0.9%	97.2 0.0%	97.3 -0.2%	97.5 -0.4%	Short	-
3M SONIA	SFI1 Comdty	96.1	Short	96.2 0.0%	96.3 -0.1%	96.2 -0.1%	96.0 0.1%	96.2 -0.1%	96.2 -0.1%	96.2 -0.1%	Short	96.2 0.0%	96.3 -0.1%	96.3 -0.1%	96.0 0.1%	96.2 0.0%	96.2 -0.1%	96.2 -0.1%	Short	-

Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

Bram Kaplan, CFA [AC] (1-212) 272-1215
bram.kaplan@jpmorgan.com
J.P. Morgan Securities LLC

Daniel Motoc, CFA [AC] (1-212) 622-0105
daniel.motoc@jpmchase.com

Arda Sebuktekin (1-212) 270-4397
arda.sebuktekin@jpmchase.com

Tony SK Lee (852) 2800-8857
tony.sk.lee@jpmorgan.com

Table 9: Commodity CTA Signals

"Current" levels are as of morning ET Oct 26; see Appendix for definitions

				Current CTA Levels								CTA Levels in 1w								
Energy	**Ticker**	**Current**	**Prior Week CTA Signal**	**1m Mom.**	**3m Mom.**	**6m Mom.**	**12m Mom.**	**50d MA**	**100d MA**	**200d MA**	**CTA Signals**	**1m Mom.**	**3m Mom.**	**6m Mom.**	**12m Mom.**	**50d MA**	**100d MA**	**200d MA**	**CTA Signals**	**1w Chg**
Brent Crude	CO1 Comdty	115.4	Long	103.8 11.2%	90.0 28.3%	113.4 1.8%	64.8 78.2%	101.0 14.3%	92.1 25.3%	91.6 25.9%	Long	100.4 15.0%	81.8 41.0%	100.9 14.4%	63.8 81.0%	101.2 14.1%	92.2 25.2%	91.8 25.7%	Long	-
WTI	CL1 Comdty	101.4	Long	90.8 11.7%	84.2 20.4%	104.6 -3.0%	60.4 67.9%	92.9 9.2%	86.4 17.4%	85.6 18.6%	Long	89.1 13.9%	76.9 31.9%	95.1 6.7%	59.9 69.5%	93.6 8.4%	86.6 17.2%	85.8 18.2%	Long	↑
US Natural Gas	NG1 Comdty	2.55	Long	3.01 -15.3%	2.74 -7.0%	2.73 -6.6%	3.56 -28.4%	2.95 -13.7%	2.98 -14.4%	3.08 -17.2%	Short	3.13 -18.5%	2.66 -4.3%	2.75 -7.3%	4.31 -40.8%	2.98 -14.5%	2.99 -14.6%	3.09 -17.4%	Short	-
Gasoline	XB1 Comdty	323.3	Neutral	335.1 -3.5%	330.1 -2.1%	370.3 -12.7%	196.7 64.3%	331.0 -2.3%	320.8 0.8%	299.3 8.0%	Neutral	325.1 -0.6%	292.1 10.7%	348.1 -7.1%	193.3 67.3%	331.3 -2.4%	321.0 0.7%	300.1 7.7%	Neutral	↑
Heating Oil	HO1 Comdty	499.4	Long	487.0 2.5%	423.4 18.0%	409.5 22.0%	242.4 106.0%	472.0 5.8%	421.9 18.4%	381.7 30.8%	Long	457.9 9.1%	386.0 29.4%	383.4 30.3%	245.8 103.2%	473.4 5.5%	423.0 18.1%	383.2 30.3%	Long	-
Low Sulphur Gasoil	QS1 Comdty	1,511	Long	1,435 5.3%	1,309 15.4%	1,306 15.7%	717 110.7%	1,414 6.8%	1,251 20.7%	1,148 31.6%	Long	1,378 9.6%	1,180 28.0%	1,174 28.7%	751 101.3%	1,417 6.6%	1,255 20.4%	1,152 31.2%	Long	-
TTF Nat Gas	TZT1 Comdty	95.55	Long	71.87 32.9%	59.22 61.3%	46.20 106.8%	31.47 203.6%	75.92 25.8%	63.63 50.2%	53.88 77.3%	Long	75.76 26.1%	54.57 75.1%	43.87 117.8%	31.97 198.8%	76.37 25.1%	63.97 49.4%	54.11 76.6%	Long	-
Base Metals	**Ticker**	**Current**	**Prior Week CTA Signal**	**1m Mom.**	**3m Mom.**	**6m Mom.**	**12m Mom.**	**50d MA**	**100d MA**	**200d MA**	**CTA Signals**	**1m Mom.**	**3m Mom.**	**6m Mom.**	**12m Mom.**	**50d MA**	**100d MA**	**200d MA**	**CTA Signals**	**1w Chg**
Copper	HG1 Comdty	653.5	Long	659.7 -0.9%	640.5 2.0%	591.2 10.5%	518.0 26.2%	658.9 -0.8%	647.2 1.0%	620.3 5.4%	Long	664.7 -1.7%	667.6 -2.1%	617.1 5.9%	496.7 31.6%	659.3 -0.9%	647.5 0.9%	620.7 5.3%	Neutral	↓
Nickel	LN1 Comdty	15,660	Short	15,878 -1.4%	17,076 -8.3%	19,220 -18.5%	15,141 3.4%	16,143 -3.0%	16,545 -5.3%	17,191 -8.9%	Short	15,546 0.7%	16,789 -6.7%	18,903 -17.2%	14,872 5.3%	16,076 -2.6%	16,494 -5.1%	17,169 -8.8%	Short	↑
Aluminum	LA1 Comdty	3,064	Short	3,220 -4.9%	3,199 -4.2%	3,558 -13.9%	2,878 6.5%	3,196 -4.1%	3,224 -5.0%	3,310 -7.4%	Short	3,123 -1.9%	3,257 -5.9%	3,563 -14.0%	2,841 7.8%	3,190 -3.9%	3,217 -4.7%	3,309 -7.4%	Short	-
Precious Metals	**Ticker**	**Current**	**Prior Week CTA Signal**	**1m Mom.**	**3m Mom.**	**6m Mom.**	**12m Mom.**	**50d MA**	**100d MA**	**200d MA**	**CTA Signals**	**1m Mom.**	**3m Mom.**	**6m Mom.**	**12m Mom.**	**50d MA**	**100d MA**	**200d MA**	**CTA Signals**	**1w Chg**
Gold	GC1 Comdty	3,994	Short	4,147 -3.7%	4,062 -1.7%	4,612 -13.4%	3,989 0.1%	4,307 -7.3%	4,235 -5.7%	4,515 -11.6%	Short	4,134 -3.4%	4,276 -6.6%	4,712 -15.2%	3,981 0.3%	4,308 -7.3%	4,235 -5.7%	4,515 -11.5%	Short	-
Silver	SI1 Comdty	53.72	Short	60.98 -11.9%	58.22 -7.7%	73.68 -27.1%	47.95 12.0%	63.19 -15.0%	62.62 -14.2%	71.75 -25.1%	Short	61.06 -12.0%	62.46 -14.0%	78.97 -32.0%	47.75 12.5%	63.12 -14.9%	62.50 -14.0%	71.66 -25.0%	Short	-
Agriculture	**Ticker**	**Current**	**Prior Week CTA Signal**	**1m Mom.**	**3m Mom.**	**6m Mom.**	**12m Mom.**	**50d MA**	**100d MA**	**200d MA**	**CTA Signals**	**1m Mom.**	**3m Mom.**	**6m Mom.**	**12m Mom.**	**50d MA**	**100d MA**	**200d MA**	**CTA Signals**	**1w Chg**
Soybean	S 1 Comdty	1,236	Long	1,293 -4.4%	1,176 5.1%	1,184 4.3%	1,083 14.1%	1,277 -3.2%	1,219 1.3%	1,183 4.4%	Long	1,294 -4.5%	1,155 7.0%	1,183 4.4%	1,107 11.6%	1,279 -3.4%	1,220 1.3%	1,184 4.3%	Long	-
Corn	C 1 Comdty	452.5	Long	515.3 -12.2%	445.2 1.6%	466.5 -3.0%	432.1 4.7%	500.4 -9.6%	466.8 -3.1%	456.2 -0.8%	Short	502.4 -9.9%	438.3 3.3%	453.9 -0.3%	431.8 4.8%	501.1 -9.7%	467.4 -3.2%	456.4 -0.9%	Short	-
Wheat	W 1 Comdty	622	Neutral	686 -9.4%	655 -5.0%	630 -1.4%	529 17.6%	696 -10.7%	660 -5.9%	622 -0.1%	Short	694 -10.5%	638 -2.5%	605 2.7%	547 13.7%	698 -11.0%	662 -6.1%	623 -0.3%	Short	↑

Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

Table 10: FX/Crypto CTA Signals

"Current" levels are as of morning ET Oct 26; see Appendix for definitions

				Current CTA Levels								CTA Levels in 1w								
FX/Crypto	**Ticker**	**Current**	**Prior Week CTA Signal**	**1m Mom.**	**3m Mom.**	**6m Mom.**	**12m Mom.**	**50d MA**	**100d MA**	**200d MA**	**CTA Signals**	**1m Mom.**	**3m Mom.**	**6m Mom.**	**12m Mom.**	**50d MA**	**100d MA**	**200d MA**	**CTA Signals**	**1w Chg**
EUR/USD	EC1 Curncy	1.12	Short	1.14 -1.6%	1.15 -2.8%	1.17 -4.5%	1.16 -3.7%	1.15 -2.2%	1.15 -2.3%	1.16 -3.4%	Short	1.13 -0.4%	1.16 -3.1%	1.18 -4.9%	1.15 -2.8%	1.14 -2.1%	1.15 -2.3%	1.16 -3.3%	Short	-
GBP/USD	BP1 Curncy	133.4	Short	132.6 0.7%	134.4 -0.7%	135.5 -1.5%	132.0 1.1%	134.0 -0.5%	133.9 -0.3%	134.3 -0.7%	Short	132.4 0.8%	134.8 -1.0%	136.0 -1.9%	130.6 2.1%	133.9 -0.4%	133.9 -0.3%	134.3 -0.6%	Short	-
JPY/USD	JY1 Curncy	63.8	Neutral	64.0 -0.3%	62.6 1.9%	63.6 0.3%	65.7 -2.9%	63.7 0.2%	63.0 1.2%	63.3 0.7%	Long	63.6 0.2%	63.5 0.4%	64.1 -0.4%	65.3 -2.4%	63.6 0.2%	63.0 1.2%	63.3 0.7%	Long	-
CAD/USD	CD1 Curncy	71.4	Short	70.7 1.0%	71.5 -0.1%	73.6 -2.9%	71.8 -0.6%	71.5 -0.2%	71.4 0.0%	72.2 -1.1%	Short	70.5 1.3%	71.6 -0.3%	73.4 -2.7%	71.0 0.5%	71.5 -0.1%	71.4 0.1%	72.2 -1.1%	Neutral	↑
AUD/USD	AD1 Curncy	70.2	Short	69.8 0.5%	70.1 0.1%	71.7 -2.1%	65.8 6.7%	70.7 -0.7%	70.3 -0.2%	70.4 -0.3%	Neutral	69.7 0.7%	70.5 -0.5%	72.3 -3.0%	64.9 8.0%	70.6 -0.7%	70.3 -0.2%	70.4 -0.3%	Short	↓
USD/CNH	XUC1 Curncy	6.66	Short	6.70 -0.6%	6.75 -1.3%	6.83 -2.5%	7.09 -6.1%	6.70 -0.6%	6.73 -1.1%	6.80 -2.0%	Short	6.70 -0.6%	6.74 -1.3%	6.80 -2.1%	7.12 -6.5%	6.70 -0.6%	6.73 -1.1%	6.79 -2.0%	Short	-
CHF/USD	SF1 Curncy	121.7	Short	121.0 0.6%	124.2 -2.0%	128.0 -4.9%	125.9 -3.3%	122.7 -0.8%	123.6 -1.5%	125.9 -3.3%	Short	121.2 0.4%	124.1 -1.9%	129.0 -5.7%	124.1 -1.9%	122.6 -0.7%	123.6 -1.5%	125.9 -3.3%	Short	-
USD/BRL	UC1 Curncy	5,598	Short	5,205 7.5%	5,084 10.1%	4,997 12.0%	5,368 4.3%	5,138 8.9%	5,142 8.9%	5,154 8.6%	Long	5,017 11.6%	5,133 9.1%	4,941 13.3%	5,400 3.7%	5,138 9.0%	5,142 8.9%	5,153 8.6%	Long	-
Bitcoin	BTC1 Curncy	79,655	Long	83,885 -5.0%	63,648 25.1%	77,097 3.3%	110,272 -27.8%	79,878 -0.3%	71,684 11.1%	73,559 8.3%	Neutral	85,162 -6.5%	64,958 22.6%	80,822 -1.4%	102,160 -22.0%	80,205 -0.7%	71,822 10.9%	73,504 8.4%	Neutral	↓
Ether	DCR1 Curncy	2,306	Neutral	2,688 -14.2%	1,890 22.0%	2,275 1.3%	3,878 -40.6%	2,482 -7.1%	2,136 8.0%	2,210 4.3%	Neutral	2,663 -13.4%	1,921 20.0%	2,328 -1.0%	3,334 -30.8%	2,490 -7.4%	2,140 7.7%	2,206 4.5%	Neutral	↓

Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

Bram Kaplan, CFA AC
(1-212) 272-1215
bram.kaplan@jpmorgan.com

J.P.Morgan

CFTC Futures Positioning

Table 11: Commitment of Traders (COT) Legacy Report

Data as of 10/20/2026

Equity	Non-Commercial	Non-Commercial (z)	Non-Commercial w/w Chg	Commercial	Commercial (z)	Commercial w/w Chg
S&P 500	-142,671	-0.5	-1,676 (0.0z)	33,922	0.5	-745 (0.0z)
Nasdaq 100	50,726	1.4	2,203 (0.2z)	-63,576	-1.4	-3,749 (-0.3z)
DJIA	6,725	0.5	-1,411 (-0.5z)	-8,113	-0.2	4,344 (1.1z)
MSCI EM	25,825	-1.0	8,313 (0.5z)	-38,998	1.3	253 (-0.1z)
Russell 2000	-41,909	-0.5	19,094 (1.6z)	34,521	0.5	-13,857 (-1.1z)
MSCI EAFE	-5,793	-0.9	-7,264 (-0.7z)	-675	0.9	7,117 (0.6z)
S&P MidCap 400	6,175	1.7	757 (0.6z)	-10,223	-1.7	-1,291 (-1.0z)
VIX	-96,509	-1.4	-54 (0.0z)	98,577	1.5	5,403 (0.5z)
Fixed Income	**Non-Commercial**	**Non-Commercial (z)**	**Non-Commercial w/w Chg**	**Commercial**	**Commercial (z)**	**Commercial w/w Chg**
2y UST	-856,783	1.6	54,796 (0.6z)	815,655	-1.5	-60,753 (-0.7z)
5y UST	-1,074,109	1.2	6,095 (-0.2z)	1,038,989	-1.3	-12,230 (0.1z)
10y UST	-964,293	-1.5	23,809 (0.4z)	930,819	1.4	-11,747 (-0.2z)
UST Ultra 10y Note	-149,347	0.4	63,134 (1.6z)	193,616	-1.2	-81,355 (-2.0z)
UST Long Bond	-68,419	0.3	58,019 (2.0z)	20,686	0.3	-56,736 (-1.6z)
UST Ultra Bond	-375,161	-2.1	-53,564 (-2.2z)	366,826	2.3	52,197 (2.0z)
Commodities	**Non-Commercial**	**Non-Commercial (z)**	**Non-Commercial w/w Chg**	**Commercial**	**Commercial (z)**	**Commercial w/w Chg**
WTI	89,325	-0.6	-11,018 (-0.7z)	-116,158	0.6	9,177 (0.5z)
US Natural Gas	-241,272	-2.1	-7,694 (-0.5z)	222,969	2.0	4,131 (0.2z)
Gasoline	69,589	0.1	-5,872 (-0.9z)	-81,044	-0.1	8,945 (1.2z)
Copper	57,397	-0.5	-11,123 (-1.6z)	-64,296	0.8	12,341 (1.6z)
Gold	199,865	0.2	-8,014 (-0.7z)	-235,582	-0.2	6,949 (0.6z)
Silver	19,795	-0.3	-2,836 (-0.9z)	-37,176	1.3	2,266 (0.7z)
Soybean	246,596	0.9	-24,503 (-0.9z)	-221,638	-0.9	24,101 (0.9z)
Corn	286,689	0.5	-71,678 (-1.3z)	-261,080	-0.6	65,878 (1.3z)
Wheat	-29,270	0.1	-11,664 (-0.9z)	29,782	-0.1	14,906 (1.1z)
Interest Rates	**Non-Commercial**	**Non-Commercial (z)**	**Non-Commercial w/w Chg**	**Commercial**	**Commercial (z)**	**Commercial w/w Chg**
3M SOFR	-2,415,267	-0.9	19,123 (0.4z)	2,414,401	0.9	-19,589 (-0.4z)
1M Fed Funds	307,293	2.1	227,102 (2.6z)	-314,872	-2.1	-223,556 (-2.7z)
FX/Crypto	**Non-Commercial**	**Non-Commercial (z)**	**Non-Commercial w/w Chg**	**Commercial**	**Commercial (z)**	**Commercial w/w Chg**
EUR/USD	-134,524	-2.0	3,049 (0.4z)	104,895	1.9	-6,330 (-0.5z)
GBP/USD	-80,420	-0.9	26,544 (2.0z)	95,508	1.1	-25,936 (-1.7z)
JPY/USD	69,878	1.5	19,671 (0.6z)	-71,087	-1.5	-22,500 (-0.7z)
CAD/USD	-74,709	0.1	20,056 (1.0z)	86,848	0.0	-18,765 (-0.8z)
AUD/USD	-127,292	-2.0	-9,993 (-0.7z)	114,925	2.0	8,555 (0.5z)
CHF/USD	-18,150	2.4	3,928 (1.1z)	34,017	-1.3	-2,255 (-0.6z)
Bitcoin	2,418	0.5	-116 (-0.3z)	-2,725	-1.8	10 (0.1z)
Ether	1,947	-1.1	-460 (-0.6z)	-2,227	1.2	387 (0.6z)

Source: J.P. Morgan Equity Derivatives Strategy, CFTC, Bloomberg Finance L.P.

Bram Kaplan, CFA AC
(1-212) 272-1215
bram.kaplan@jpmorgan.com

J.P.Morgan

Table 12: Traders in Financial Futures and COT Disaggregated Report

Data as of 10/20/2026

Equity	Asset Manager	Asset Manager (z)	Asset Manager w/w Chg	Leveraged Funds	Leveraged Funds (z)	Leveraged Funds w/w Chg	Dealer	Dealer (z)	Dealer w/w Chg	Other Reportables	Other Reportables (z)	Other Reportables
S&P 500	901,377	-1.1	-14,458 (-0.5z)	-371,245	0.2	37,481 (0.7z)	-636,931	1.0	-34,573 (-0.7z)	-1,950	-0.9	9,129 (0.8z)
Nasdaq 100	77,790	0.4	2,467 (0.4z)	950	1.3	12,543 (1.0z)	-94,122	-1.2	-16,190 (-1.1z)	2,532	-1.2	-366 (-0.1z)
DJIA	13,066	1.1	2,193 (0.5z)	-1,967	0.9	-4,776 (-1.6z)	-13,153	-1.3	5,296 (1.4z)	666	2.1	220 (0.8z)
MSCI EM	505,642	0.5	17,674 (0.6z)	22,506	-0.4	-7,083 (-0.2z)	-530,274	0.4	-8,627 (-0.5z)	-11,047	-0.1	6,602 (0.7z)
Russell 2000	30,960	0.5	1,432 (0.1z)	-53,035	0.4	38,612 (3.0z)	23,003	-0.4	-34,500 (-2.8z)	-8,316	-0.7	-307 (-0.1z)
MSCI EAFE	268,541	2.0	15,575 (1.5z)	-7,630	-0.2	-6,031 (-0.6z)	-266,151	-1.2	-9,687 (-0.9z)	-1,228	-0.8	-4 (0.0z)
S&P MidCap 400	6,425	1.2	-118 (-0.1z)	5,095	1.4	-1,179 (-1.0z)	-15,240	-2.3	763 (0.7z)	-328	-0.8	0 (0.0z)
VIX	-57,459	-1.4	19,836 (2.1z)	-20,890	0.4	-21,995 (-1.9z)	78,379	2.0	8,331 (0.8z)	2,038	0.9	-823 (-0.7z)
Fixed Income	**Asset Manager**	**Asset Manager (z)**	**Asset Manager w/w Chg**	**Leveraged Funds**	**Leveraged Funds (z)**	**Leveraged Funds w/w Chg**	**Dealer**	**Dealer (z)**	**Dealer w/w Chg**	**Other Reportables**	**Other Reportables (z)**	**Other Reportables**
2y UST	1,723,580	-1.2	-72,286 (-0.7z)	-1,285,548	1.6	81,860 (0.8z)	-494,117	-1.0	-12,396 (-0.2z)	14,957	-2.2	-3,135 (-0.1z)
5y UST	2,928,052	-0.6	-56,900 (-0.5z)	-2,117,544	1.0	-13,153 (-0.3z)	-850,470	-1.1	67,574 (1.1z)	4,842	-1.1	-3,656 (-0.1z)
10y UST	2,925,416	2.4	26,854 (0.1z)	-2,107,224	-0.1	98,713 (1.3z)	-682,228	-1.7	-49,415 (-0.7z)	-169,438	-2.8	-64,090 (-1.2z)
UST Ultra 10y Note	737,266	1.3	-60,228 (-1.5z)	-318,477	0.3	17,381 (0.4z)	-309,570	-1.1	11,622 (0.5z)	-64,950	-3.1	13,004 (0.8z)
UST Long Bond	388,007	-1.5	-3,527 (-0.1z)	-162,757	2.1	-8,737 (-0.4z)	-182,706	1.3	-3,407 (-0.2z)	-90,277	-2.2	16,954 (2.0z)
UST Ultra Bond	849,383	-1.5	72,891 (1.9z)	-804,808	-0.1	-89,079 (-2.9z)	-93,420	2.4	4,145 (0.1z)	40,510	1.9	10,676 (1.5z)
Commodities	**Producer**	**Producer (z)**	**Producer w/w Chg**	**Managed Money**	**Managed Money (z)**	**Managed Money w/w Chg**	**Swap Dealer**	**Swap Dealer (z)**	**Swap Dealer w/w Chg**	**Other Reportables**	**Other Reportables (z)**	**Other Reportables**
Brent Crude	-389,668	-0.1	-21,830 (-0.5z)	221,344	0.0	14,414 (0.3z)	285,875	-0.9	113 (0.1z)	-126,035	0.8	10,769 (0.3z)
WTI	359,568	1.0	6,425 (0.3z)	36,187	-0.7	-4,387 (-0.4z)	-475,726	-1.4	2,752 (0.3z)	53,138	-1.2	-6,631 (-0.4z)
US Natural Gas	-13,154	-0.1	-2,273 (-0.2z)	-157,409	-2.1	-30,928 (-1.2z)	236,123	1.9	6,404 (0.6z)	-83,863	0.4	23,234 (1.2z)
Gasoline	-83,962	0.6	9,894 (1.5z)	93,712	1.5	-4,196 (-0.7z)	2,918	-0.8	-949 (-0.2z)	-24,123	-1.4	-1,676 (-0.3z)
Copper	-77,406	1.4	13,060 (1.8z)	47,690	-1.4	-13,368 (-1.9z)	13,110	-0.6	-719 (-0.2z)	9,707	1.3	2,245 (0.5z)
Gold	-14,783	1.0	3,654 (0.6z)	104,206	-0.5	-2,883 (-0.4z)	-220,799	-1.8	3,295 (0.4z)	95,659	0.8	-5,131 (-0.7z)
Silver	-15,029	0.8	772 (0.4z)	3,505	-1.8	-2,114 (-0.7z)	-22,147	1.5	1,494 (0.6z)	16,290	0.9	-722 (-0.3z)
Soybean	-287,435	-0.3	15,543 (0.6z)	221,325	0.9	-28,482 (-1.0z)	65,797	-2.3	8,558 (1.4z)	25,271	-0.4	3,979 (0.5z)
Corn	-528,744	-0.4	71,745 (1.3z)	164,423	0.3	-82,002 (-1.5z)	267,664	-1.7	-5,867 (-0.7z)	122,266	1.1	10,324 (0.8z)
Wheat	-40,646	-0.1	14,379 (1.1z)	-52,005	-0.3	-19,450 (-1.3z)	70,428	0.2	527 (0.1z)	22,735	1.1	7,786 (1.7z)
Interest Rates	**Asset Manager**	**Asset Manager (z)**	**Asset Manager w/w Chg**	**Leveraged Funds**	**Leveraged Funds (z)**	**Leveraged Funds w/w Chg**	**Dealer**	**Dealer (z)**	**Dealer w/w Chg**	**Other Reportables**	**Other Reportables (z)**	**Other Reportables**
3M SOFR	-231,610	-1.2	235,309 (1.8z)	-2,373,040	-0.9	-127,533 (-0.5z)	2,628,838	0.7	-92,122 (-0.6z)	-25,054	0.1	-16,120 (-0.5z)
1M Fed Funds	-97,802	-1.0	-75,723 (-1.9z)	376,169	1.8	231,272 (2.9z)	-302,239	-1.4	-149,309 (-2.3z)	16,293	-0.4	-2,694 (-0.3z)
FX/Crypto	**Asset Manager**	**Asset Manager (z)**	**Asset Manager w/w Chg**	**Leveraged Funds**	**Leveraged Funds (z)**	**Leveraged Funds w/w Chg**	**Dealer**	**Dealer (z)**	**Dealer w/w Chg**	**Other Reportables**	**Other Reportables (z)**	**Other Reportables**
EUR/USD	190,421	-1.6	-9,497 (-0.4z)	-68,840	-1.8	-839 (0.1z)	-155,718	1.7	9,045 (0.1z)	4,508	-1.1	-1,990 (-1.0z)
GBP/USD	-134,621	-1.2	11,114 (0.9z)	24,104	-0.1	13,791 (1.9z)	125,120	1.1	-25,676 (-1.4z)	485	0.3	1,379 (1.9z)
JPY/USD	79,103	1.4	13,284 (0.9z)	-22,890	1.4	10,783 (0.5z)	-99,290	-1.2	-25,479 (-0.7z)	41,868	-0.9	-1,417 (-0.1z)
CAD/USD	-5,399	0.4	25,405 (1.4z)	-91,113	-1.4	-3,001 (-0.3z)	100,798	0.2	-20,183 (-0.8z)	7,853	-1.4	-930 (-0.7z)
AUD/USD	-128,739	-2.4	-12,819 (-1.1z)	52,497	0.6	1,906 (0.1z)	60,274	1.6	11,373 (0.7z)	3,601	-0.7	-1,898 (-1.3z)
CHF/USD	-25,535	2.2	5,921 (1.4z)	-13,607	-1.4	1,334 (0.8z)	54,643	0.0	-4,871 (-1.1z)	366	0.3	-711 (-1.4z)
Bitcoin	2,999	-1.0	-643 (-1.1z)	-4,942	1.6	902 (0.6z)	1,392	-2.4	-449 (-0.7z)	244	-0.4	84 (0.1z)
Ether	-1,306	-1.5	-635 (-0.7z)	-5,128	1.1	1,155 (0.3z)	7,550	-1.0	-750 (-0.3z)	-1,396	-2.0	157 (0.5z)

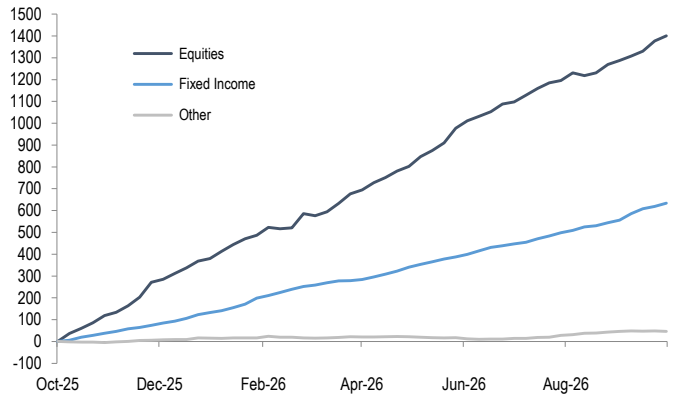
Source: J.P. Morgan Equity Derivatives Strategy, CFTC, Bloomberg Finance L.P.

Bram Kaplan, CFA [AC] (1-212) 272-1215
bram.kaplan@jpmorgan.com
J.P. Morgan Securities LLC
Daniel Motoc, CFA [AC] (1-212) 622-0105
daniel.motoc@jpmchase.com
Arda Sebuktekin (1-212) 270-4397
arda.sebuktekin@jpmchase.com
Tony SK Lee (852) 2800-8857
tony.sk.lee@jpmorgan.com

ETF Flows

Figure 1: Asset Class Flows

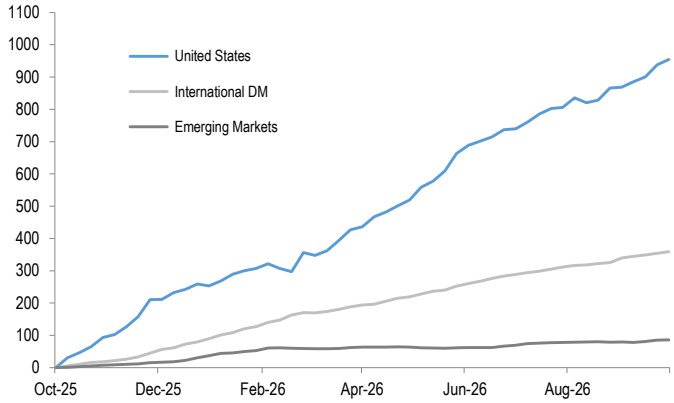
$Bn



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

Figure 2: Equity Regional Flows

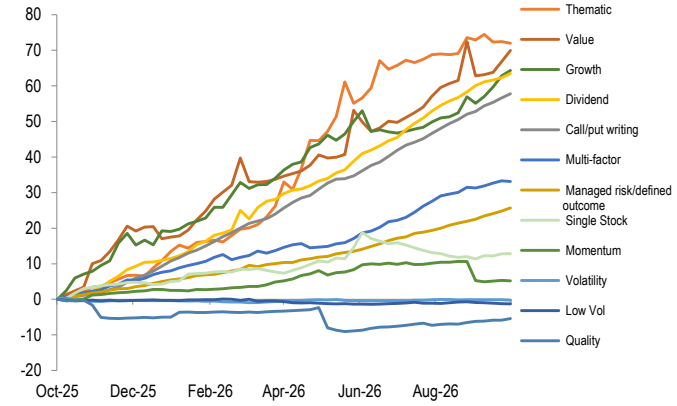
$Bn



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

Figure 3: Equity Style Flows

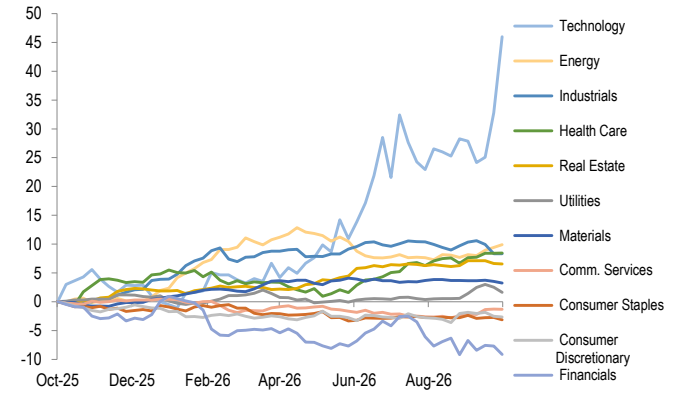
$Bn



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

Figure 4: Equity Sector Flows

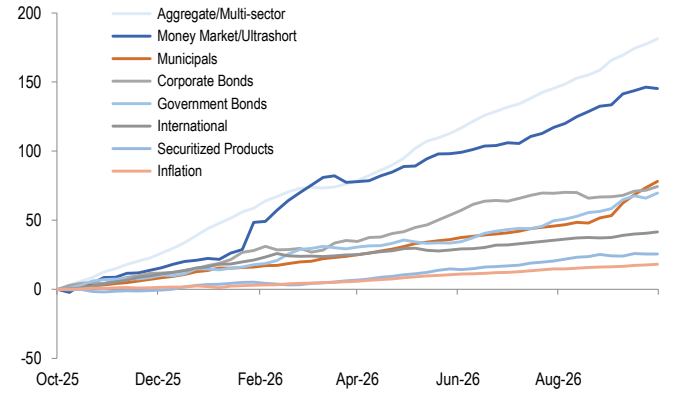
$Bn



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

Figure 5: Fixed Income Flows

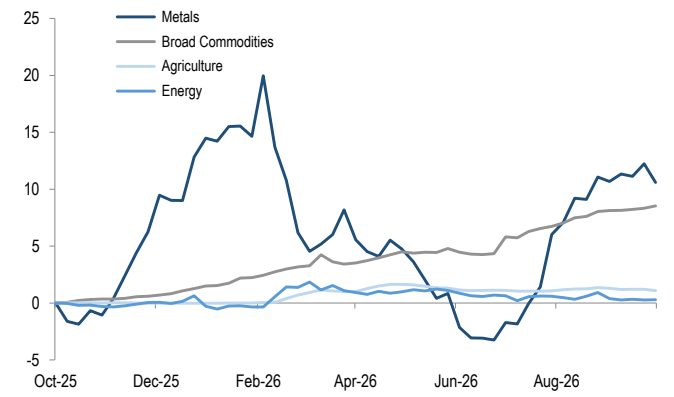
$Bn



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

Figure 6: Commodity Flows

$Bn



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

Bram Kaplan, CFA AC (1-212) 272-1215
bram.kaplan@jpmorgan.com
J.P. Morgan Securities LLC
Daniel Motoc, CFA AC (1-212) 622-0105
daniel.motoc@jpmchase.com

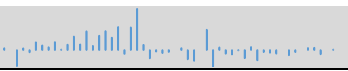
Arda Sebuktekin (1-212) 270-4397
arda.sebuktekin@jpmchase.com
Tony SK Lee (852) 2800-8857
tony.sk.lee@jpmorgan.com

Table 13: ETF Flows by Asset Class

Asset Class	Net Flow - 1y History	Net 1w Flow ($Bn)	Net 4w Flow ($Bn)	Net 1w Flow (z)	Net 4w Flow (z)
Equities		23.4	112.6	-0.2	0.1
Fixed Income		15.2	78.0	0.6	2.4
Commodities		-1.6	0.0	-0.9	-0.4
Currency/Multi-asset		0.0	-0.2	-0.4	-0.7

Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

Table 14: Equity ETF Flows by Region

Region	Net Flow - 1y History	Net 1w Flow ($Mn)	Net 4w Flow ($Mn)	Net 1w Flow (z)	Net 4w Flow (z)
United States		16,993	86,061	-0.1	0.4
International DM		5,352	19,445	-0.5	-0.2
Europe (Developed)		-121	-275	-0.5	-0.7
Asia Pac (Developed)		405	1,471	0.3	0.3
Japan		382	1,133	0.3	0.2
Canada		246	707	1.4	1.5
Emerging Markets		1,092	7,105	-0.2	0.1
Latam		-133	3,815	-0.5	2.5
Mexico		7	-86	0.4	-0.2
Brazil		72	4,302	-0.1	3.5
EM Asia		-39	641	-0.2	0.1
China		-91	-894	0.0	-0.7
Korea		45	1,390	-0.2	0.2
India		-7	-203	0.3	0.1
CEEMEA		45	192	0.7	1.0
MENA		2	-4	-0.2	-0.4

Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

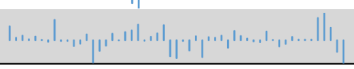
Bram Kaplan, CFA AC (1-212) 272-1215
bram.kaplan@jpmorgan.com
J.P. Morgan Securities LLC
Daniel Motoc, CFA AC (1-212) 622-0105
daniel.motoc@jpmchase.com
Arda Sebuktekin (1-212) 270-4397
arda.sebuktekin@jpmchase.com
Tony SK Lee (852) 2800-8857
tony.sk.lee@jpmorgan.com

Table 15: Equity ETF Flows by Style

Style	Net Flow - 1y History	Net 1w Flow ($Mn)	Net 4w Flow ($Mn)	Net 1w Flow (z)	Net 4w Flow (z)
Growth		1,516	9,242	0.1	1.3
Value		3,124	7,119	0.5	0.4
Dividend		1,188	3,431	0.0	-1.0
Multi-factor		-207	1,791	-1.4	-0.6
Momentum		-125	-59	-0.3	-0.2
Quality		410	750	0.5	0.5
Low Vol		-57	-423	-0.2	-1.2
Call/put writing		1,195	4,952	0.2	0.4
Managed risk/defined outcome		853	3,176	1.3	2.2
Volatility		-86	-121	-0.8	-0.5
Thematic		-474	-900	-0.7	-1.3
Single Stock		90	1,493	-0.2	0.2

Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

Table 16: Equity ETF Flows by Sector

Sector	Net Flow - 1y History	Net 1w Flow ($Mn)	Net 4w Flow ($Mn)	Net 1w Flow (z)	Net 4w Flow (z)
Consumer Discretionary		-124	-824	-0.2	-0.9
Consumer Staples		-402	-756	-1.0	-1.0
Energy		474	1,712	0.4	0.5
Financials		-1,457	-2,428	-1.1	-0.7
Health Care		31	799	-0.2	0.1
Industrials		-3	-2,043	-0.3	-2.0
Materials		-309	-403	-1.4	-1.1
Real Estate		-87	-580	-0.6	-1.5
Technology		13,190	18,110	3.5	3.1
Comm. Services		-42	843	-0.1	1.5
Utilities		-901	185	-2.4	0.0

Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

Bram Kaplan, CFA AC (1-212) 272-1215
bram.kaplan@jpmorgan.com
J.P. Morgan Securities LLC
Daniel Motoc, CFA AC (1-212) 622-0105
daniel.motoc@jpmchase.com
Arda Sebuktekin (1-212) 270-4397
arda.sebuktekin@jpmchase.com
Tony SK Lee (852) 2800-8857
tony.sk.lee@jpmorgan.com

J.P.Morgan

Table 17: Fixed Income ETF Flows by Segment

Segment	Net Flow - 1y History	Net 1w Flow ($Bn)	Net 4w Flow ($Bn)	Net 1w Flow (z)	Net 4w Flow (z)
Aggregate/Multi-sector		3.8	15.4	0.2	0.3
Government Bonds		3.7	11.4	1.5	1.7
Short-term		0.6	4.3	0.1	0.8
Medium-term		2.8	4.4	2.1	1.0
Long-term		0.1	1.7	0.0	0.4
Corporate Bonds		2.7	7.4	0.7	0.4
IG		1.8	6.7	0.5	0.6
HY		0.8	0.5	0.7	-0.1
Money Market/Ultrashort		-0.9	11.8	-1.1	0.0
Municipals		4.4	24.8	1.8	4.1
Mortgage		0.2	-1.6	0.3	-0.5
Inflation		0.5	1.9	0.5	0.7
International		1.1	3.9	0.3	0.3
Loans		-0.2	3.0	-1.2	0.8

Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

Table 18: Commodity ETF Flows by Segment

Segment	Net Flow - 1y History	Net 1w Flow ($Mn)	Net 4w Flow ($Mn)	Net 1w Flow (z)	Net 4w Flow (z)
Broad Commodities		196	405	0.1	-0.5
Energy		6	-111	0.0	-0.2
Precious Metals		-1,646	-31	-0.9	-0.2
Base Metals		-7	-52	-0.3	-0.6
Agriculture		-115	-205	-1.3	-1.0

Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

Bram Kaplan, CFA [AC] (1-212) 272-1215
bram.kaplan@jpmorgan.com
J.P. Morgan Securities LLC
Daniel Motoc, CFA [AC] (1-212) 622-0105
daniel.motoc@jpmchase.com
Arda Sebuktekin (1-212) 270-4397
arda.sebuktekin@jpmchase.com
Tony SK Lee (852) 2800-8857
tony.sk.lee@jpmorgan.com

Table 19: Currency/Multi-asset ETF Flows by Segment

Currency/Multi-asset	Net Flow - 1y History	Net 1w Flow ($Mn)	Net 4w Flow ($Mn)	Net 1w Flow (z)	Net 4w Flow (z)
Fiat		-8	25	-0.2	0.0
Cryptocurrency		71	-1,778	0.0	-0.7
Asset Allocation		89	469	-0.4	-0.6
Liquid Alts		-176	1,061	-1.5	0.0

Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

Bram Kaplan, CFA AC (1-212) 272-1215
bram.kaplan@jpmorgan.com
J.P. Morgan Securities LLC
Daniel Motoc, CFA AC (1-212) 622-0105
daniel.motoc@jpmchase.com
Arda Sebuktekin (1-212) 270-4397
arda.sebuktekin@jpmchase.com
Tony SK Lee (852) 2800-8857
tony.sk.lee@jpmorgan.com

J.P.Morgan

Table 20: Top-Performing ETFs

1W & 4W, excludes levered ETFs

	ETF Name	1w Perf	Ticker	ETF Name	4w Perf
WCBR	WisdomTree Cybersecurity Fund	12.7%	BWET	Breakwave Tanker Shipping ETF	31.4%
BUG	Global X Cybersecurity ETF	12.6%	EWZS	iShares MSCI Brazil Small-Cap	20.4%
WCLD	WisdomTree Cloud Computing Fun	11.8%	FLBR	Franklin FTSE Brazil ETF	14.2%
IHAK	iShares Cybersecurity and Tech	10.9%	EWZ	iShares MSCI Brazil ETF	13.3%
HACK	Amplify Cybersecurity ETF	9.2%	IEO	iShares U.S. Oil & Gas Explora	12.1%

Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

Table 21: Bottom-Performing ETFs

1W & 4W, excludes levered ETFs

Ticker	ETF Name	1w Perf	Ticker	ETF Name	4w Perf
SGDJ	Sprott Junior Gold Miners ETF	-12.3%	WGMI	CoinShares Bitcoin Mining and	-37.2%
GDMN	Wisdomtree Efficient Gold Plus	-11.3%	STCE	Schwab Crypto Thematic Natural	-28.4%
SIL	Global X Silver Miners ETF	-10.2%	BKCH	Global X Blockchain ETF	-28.0%
UNG	United States Natural Gas Fund	-10.1%	ZCSH	The Zcash ETF	-26.5%
SLVP	iShares MSCI Global Silver and	-10.1%	SMCY	Yieldmax Smci Option Income St	-26.5%

Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

Table 22: Largest Fund Inflows

1W & 4W

Ticker	ETF Name	1w Inflow	Ticker	ETF Name	4w Inflow
SOXX	iShares Semiconductor ETF	$6.5Bn (3.9z)	VOO	Vanguard S&P 500 ETF	$23.3Bn (0.5z)
VOO	Vanguard S&P 500 ETF	$6.2Bn (0.2z)	SOXL	Direxion Daily Semiconductor B	$8.9Bn (2.8z)
SOXL	Direxion Daily Semiconductor B	$2.8Bn (1.8z)	SPYM	State Street SPDR Portfolio S&	$8.4Bn (0.7z)
SMH	VanEck Semiconductor ETF	$2.6Bn (2.1z)	MUB	iShares National Muni Bond ETF	$7.0Bn (4.0z)
MUB	iShares National Muni Bond ETF	$1.6Bn (2.8z)	SGOV	iShares 0-3 Month Treasury Bon	$4.7Bn (0.2z)

Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

Table 23: Largest Fund Outflows

1W & 4W

Ticker	ETF Name	1w Outflow	Ticker	ETF Name	4w Outflow
QQQ	Invesco QQQ Trust Series 1	$-6.5Bn (-1.8z)	SPY	State Street SPDR S&P 500 ETF	$-4.2Bn (-0.6z)
SPY	State Street SPDR S&P 500 ETF	$-5.2Bn (-0.7z)	IVV	iShares Core S&P 500 ETF	$-3.0Bn (-0.3z)
GLD	SPDR Gold Shares	$-2.3Bn (-1.5z)	DRAM	Roundhill Memory ETF	$-2.4Bn (-1.2z)
IWM	iShares Russell 2000 ETF	$-1.8Bn (-1.0z)	ARKK	ARK Innovation ETF	$-2.3Bn (-1.5z)
XLF	State Street Financial Select	$-1.6Bn (-1.5z)	XLF	State Street Financial Select	$-2.1Bn (-0.9z)

Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

Bram Kaplan, CFA [AC] (1-212) 272-1215
bram.kaplan@jpmorgan.com
J.P. Morgan Securities LLC
Daniel Motoc, CFA [AC] (1-212) 622-0105
daniel.motoc@jpmchase.com

Arda Sebuktekin (1-212) 270-4397
arda.sebuktekin@jpmchase.com
Tony SK Lee (852) 2800-8857
tony.sk.lee@jpmorgan.com

Futures Liquidity

Table 24: Equity Futures Liquidity

See Appendix for definitions

Americas	Ticker	Bid/Ask Spread (% of notional, bps)	Market Depth (contracts, top of book)	Market Depth (z)	ADV (prior week - $Bn)	ADV (prior week - z)	1y ADV ($Bn)
S&P 500	ES1 Index	0.3	22	0.0	708.5	0.8	577.3
Nasdaq 100	NQ1 Index	0.3	2	-0.9	413.5	1.4	319.1
Russell 2000	RTY1 Index	0.6	4	0.0	24.4	-0.4	28.1
DJIA	DM1 Index	0.5	2	-1.1	24.6	-0.1	25.0
S&P MidCap 400	FA1 Index	2.4	2	1.0	4.8	-0.1	5.1
MSCI EM	MES1 Index	1.0	6	-0.7	8.9	-0.4	12.9
MSCI EAFE	MFS1 Index	1.0	3	-0.4	7.0	-0.2	8.8
VIX	UX1 Index	27.6	205	0.7	4.6	0.6	3.6
S&P TSX 60	PT1 Index	1.8	3	-0.7	4.7	-0.3	6.5
Bovespa	XB1 Index	0.3	71	-0.9	163.6	1.9	122.9
EMEA	**Ticker**	**Bid/Ask Spread (% of notional, bps)**	**Market Depth (contracts, top of book)**	**Market Depth (z)**	**ADV (prior week - $Bn)**	**ADV (prior week - z)**	**1y ADV ($Bn)**
Euro Stoxx 50	VG1 Index	1.7	212	0.3	51.9	0.1	50.7
FTSE 100	Z 1 Index	0.7	8	2.7	11.7	-0.1	12.9
DAX	GX1 Index	1.0	3	1.3	24.8	0.4	21.9
SMI	SM1 Index	1.0	10	1.7	6.2	0.2	5.5
CAC 40	CF1 Index	1.0	5	1.2	4.0	-0.5	5.7
FTSE/JSE Top 40	AI1 Index	3.6	3	1.2	0.8	-0.4	1.3
BIST 30	A51 Index	2.0	11	-0.9	1.1	0.4	1.0
APAC	**Ticker**	**Bid/Ask Spread (% of notional, bps)**	**Market Depth (contracts, top of book)**	**Market Depth (z)**	**ADV (prior week - $Bn)**	**ADV (prior week - z)**	**1y ADV ($Bn)**
TOPIX	TP1 Index	1.5	9	-0.2	15.6	-0.1	16.3
Nikkei 225	NK1 Index	2.0	9	-0.4	17.0	0.2	15.4
ASX SPI 200	XP1 Index	1.4	15	2.2	6.5	-0.2	7.8
KOSPI 200	KM1 Index	1.9	2	-0.7	28.2	-0.2	29.6
Hang Seng	HI1 Index	0.7	5	2.6	13.2	-1.0	18.9
Hang Seng China Ent.	HC1 Index	1.4	30	2.5	5.4	-0.7	7.9
FTSE China A50	XU1 Index	0.9	19	0.2	6.2	-0.2	6.9
CSI 300	IFB1 Index	1.3	3	-0.6	24.6	0.2	23.8
CSI 500	FFD1 Index	1.6	3	-0.1	39.0	0.1	38.2
CSI 1000	IFD1 Index	1.4	4	1.1	67.0	1.0	54.8
NIFTY 50	JGS1 Index	1.2	3	-0.8	6.7	1.6	4.0
SET 50	BC1 Index	1.3	31	0.2	0.9	-0.3	1.0
SSE50	FFB1 Index	1.4	3	-0.7	8.0	0.7	7.1
SGX FTSE Taiwan	TWT1 Index	1.6	3	-0.8	7.2	-0.3	8.5
TAIEX	FT1 Index	1.1	2	-0.7	29.3	0.1	28.2
MSCI Singapore	QZ1 Index	1.7	4	-0.7	1.0	-0.4	1.5

Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

Bram Kaplan, CFA AC (1-212) 272-1215
bram.kaplan@jpmorgan.com
J.P. Morgan Securities LLC
Daniel Motoc, CFA AC (1-212) 622-0105
daniel.motoc@jpmchase.com

Arda Sebuktekin (1-212) 270-4397
arda.sebuktekin@jpmchase.com
Tony SK Lee (852) 2800-8857
tony.sk.lee@jpmorgan.com

Table 25: Fixed Income Futures Liquidity

See Appendix for definitions

Americas	Ticker	Bid/Ask Spread (% of notional, bps)	Market Depth (contracts, top of book)	Market Depth (z)	ADV (prior week - $Bn)	ADV (prior week - z)	1y ADV ($Bn)
2y UST	TU1 Comdty	0.4	1,516	-1.6	190.8	-0.4	255.9
5y UST	FV1 Comdty	0.8	1,295	-0.9	166.9	-0.3	201.1
10y UST	TY1 Comdty	1.5	2,730	-0.2	267.0	0.0	272.2
UST Ultra 10y Note	UXY1 Comdty	1.5	962	-0.3	83.6	-0.1	87.8
UST Long Bond	US1 Comdty	3.0	709	-0.6	65.2	0.0	65.8
UST Ultra Bond	WN1 Comdty	2.9	438	0.0	50.2	-0.1	55.6
Canada 10y	CN1 Comdty	0.9	149	2.6	19.8	0.1	18.5
EMEA	**Ticker**	**Bid/Ask Spread (% of notional, bps)**	**Market Depth (contracts, top of book)**	**Market Depth (z)**	**ADV (prior week - $Bn)**	**ADV (prior week - z)**	**1y ADV ($Bn)**
Germany 2y	DU1 Comdty	0.5	2,201	-0.9	117.0	-0.1	122.5
Germany 5y	OE1 Comdty	0.9	2,258	-0.1	136.8	-0.1	143.0
Germany 10y	RX1 Comdty	0.8	609	-0.5	172.2	-0.2	188.8
Germany 30y	UB1 Comdty	2.0	124	0.1	18.3	-0.5	24.4
10y Gilt	G 1 Comdty	1.3	180	1.6	35.4	-0.2	40.4
France 10y	OAT1 Comdty	1.0	90	-1.7	47.1	0.1	44.9
Italy 10y	IK1 Comdty	0.9	134	-0.9	44.8	-0.4	52.5
APAC	**Ticker**	**Bid/Ask Spread (% of notional, bps)**	**Market Depth (contracts, top of book)**	**Market Depth (z)**	**ADV (prior week - $Bn)**	**ADV (prior week - z)**	**1y ADV ($Bn)**
Japan 10y	JB1 Comdty	0.8	24	-0.1	34.6	-0.1	36.2
Australia 10y	XM1 Comdty	0.6	1,179	0.9	16.3	-0.2	18.7
China 10y	TFT1 Comdty	0.5	161	0.7	18.6	0.2	17.3
Korea 10y	KAA1 Comdty	1.1	53	1.3	7.8	0.5	7.0

Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

Table 26: Interest Rates Futures Liquidity

See Appendix for definitions

Global	Ticker	Bid/Ask Spread (% of notional, bps)	Market Depth (contracts, top of book)	Market Depth (z)	ADV (prior week - $Bn)	ADV (prior week - z)	1y ADV ($Bn)
3M SOFR	SFR1 Comdty	0.5	8,337	-0.6	756.3	-0.6	929.2
3M Euribor	ER1 Comdty	0.6	1,543	-0.9	532.1	-0.1	574.7
3M SONIA	SFI1 Comdty	0.6	916	-0.8	303.0	-0.2	331.6

Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

Bram Kaplan, CFA AC (1-212) 272-1215
bram.kaplan@jpmorgan.com
J.P. Morgan Securities LLC
Daniel Motoc, CFA AC (1-212) 622-0105
daniel.motoc@jpmchase.com
Arda Sebuktekin (1-212) 270-4397
arda.sebuktekin@jpmchase.com
Tony SK Lee (852) 2800-8857
tony.sk.lee@jpmorgan.com

Table 27: Commodity Futures Liquidity

See Appendix for definitions

Energy	Ticker	Bid/Ask Spread (% of notional, bps)	Market Depth (contracts, top of book)	Market Depth (z)	ADV (prior week - $Bn)	ADV (prior week - z)	1y ADV ($Bn)
Brent Crude	CO1 Comdty	1.7	4	-0.6	138.5	0.5	114.1
WTI	CL1 Comdty	1.5	6	-0.4	94.9	0.4	81.0
US Natural Gas	NG1 Comdty	4.2	13	0.4	16.0	-0.2	18.2
Gasoline	XB1 Comdty	2.8	2	-1.0	23.5	0.1	23.1
Heating Oil	HO1 Comdty	4.4	1	-1.0	27.4	0.1	26.5
Low Sulphur Gasoil	QS1 Comdty	3.9	5	-0.7	37.1	-0.1	38.6
TTF Nat Gas	TZT1 Comdty	8.1	10	-1.2	30.7	1.5	17.2
Base Metals	**Ticker**	**Bid/Ask Spread (% of notional, bps)**	**Market Depth (contracts, top of book)**	**Market Depth (z)**	**ADV (prior week - $Bn)**	**ADV (prior week - z)**	**1y ADV ($Bn)**
Copper	HG1 Comdty	1.6	3	0.1	10.8	-0.1	11.0
Nickel	LN1 Comdty	10.2	4	-0.1	5.1	-0.1	5.2
Aluminum	LA1 Comdty	15.4	3	-0.3	11.1	-0.5	13.4
Precious Metals	**Ticker**	**Bid/Ask Spread (% of notional, bps)**	**Market Depth (contracts, top of book)**	**Market Depth (z)**	**ADV (prior week - $Bn)**	**ADV (prior week - z)**	**1y ADV ($Bn)**
Gold	GC1 Comdty	0.7	2	0.9	79.1	-0.4	95.9
Silver	SI1 Comdty	2.0	2	1.0	13.9	-0.7	30.9
Agriculture	**Ticker**	**Bid/Ask Spread (% of notional, bps)**	**Market Depth (contracts, top of book)**	**Market Depth (z)**	**ADV (prior week - $Bn)**	**ADV (prior week - z)**	**1y ADV ($Bn)**
Soybean	S 1 Comdty	2.2	51	0.5	27.9	2.3	17.6
Corn	C 1 Comdty	5.6	429	0.7	11.0	0.1	10.8
Wheat	W 1 Comdty	4.3	43	-0.1	4.8	-0.2	5.1

Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

Table 28: FX/Crypto Futures Liquidity

See Appendix for definitions

FX/Crypto	Ticker	Bid/Ask Spread (% of notional, bps)	Market Depth (contracts, top of book)	Market Depth (z)	ADV (prior week - $Bn)	ADV (prior week - z)	1y ADV ($Bn)
EUR/USD	EC1 Curncy	0.5	71	2.2	25.6	-0.3	31.9
GBP/USD	BP1 Curncy	0.9	104	1.5	6.1	-0.7	8.6
JPY/USD	JY1 Curncy	0.9	84	2.1	15.4	0.1	14.6
CAD/USD	CD1 Curncy	0.8	149	2.5	7.0	0.3	6.0
AUD/USD	AD1 Curncy	0.8	79	2.0	5.4	-0.7	7.8
USD/CNH	XUC1 Curncy	0.2	39	3.1	16.4	-0.7	20.6
CHF/USD	SF1 Curncy	0.7	11	2.0	4.5	-0.2	4.9
USD/BRL	UC1 Curncy	1.2	73	1.5	13.3	0.4	12.3
Bitcoin	BTC1 Curncy	4.1	2	-0.9	4.9	0.0	4.9
Ether	DCR1 Curncy	4.5	3	0.0	1.9	-0.2	2.1

Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

Bram Kaplan, CFA AC (1-212) 272-1215
bram.kaplan@jpmorgan.com
J.P. Morgan Securities LLC
Daniel Motoc, CFA AC (1-212) 622-0105
daniel.motoc@jpmchase.com
Arda Sebuktekin (1-212) 270-4397
arda.sebuktekin@jpmchase.com
Tony SK Lee (852) 2800-8857
tony.sk.lee@jpmorgan.com

Appendix

**Unless otherwise specified, all Z-scores are calculated over a 1Y period. Z-score = (Current Value - Average)/Standard Deviation)*

Flows and Positioning Definitions

1. **Net Flows:** The net flow traded on a futures contract, calculated across the futures chain. Volumes are classified via an attribution of trade-by-trade flows, based on whether the trade price was on the bid side (identified as a sell flow) or offer side (identified as a buy flow) of the order book.

2. **AM + HF:** Asset managers' and Hedge Funds' positioning in the relevant futures contract, as reported in the latest CFTC Traders in Financial Futures report. Expressed as a Z-score.

3. **Non-Commercial:** Non-commercial entities' positioning in the relevant futures contract, as reported in the latest CFTC Commitment of Traders report. Expressed as a Z-score.

4. **OI Change:** The weekly change in the notional value of contracts outstanding for the relevant futures contracts.

5. **OI Correlation w/Spot:** The 3M correlation between changes in the number of contracts outstanding and the price of the future's underlying spot index. A positive (negative) correlation indicates likely increased (decreased) positioning as open interest is growing at the same time as the market is rallying (selling off), indicating the incremental contracts opened are most likely initiated by investors looking to go directionally long (short), respectively.

6. **CTA Signals:** Whether CTAs are likely to be short, neutral, or long a certain futures contract based on key momentum signals – see more below.

CTA Signals Definitions

1. **CTA Signal:** Whether CTAs are likely to be short, neutral, or long in a certain futures contract based on key momentum signals. Our metric looks at seven signals, assigning a score of -1, 0, or 1 depending on whether the index is below/at/above the relevant signal. The net score determines our CTA metric. The seven signals we look at are 1M, 3M, 6M, and 12M momentum and 50d, 100d, and 200d moving averages. In each cell, the top number is the price level of the signal that indicates where CTAs are likely to buy or sell the contract, and the bottom number is the distance from the signal to the current futures price. Cells shaded in green are currently long momentum signals and those in red are short signals.

2. **1w Chg:** Indicates the change in CTA signals in one week's time if futures levels are unchanged. "↑" indicates CTAs are likely to become longer/less short, "↓" indicates they will become shorter/less long, and "-" indicates no change.

Liquidity Definitions

1. **Bid/Ask Spread:** The average bid/ask spread on the futures contract, expressed as a percentage of notional, in bps.

2. **Market Depth:** The average number of contracts at the top of the order book intraday (averaged across the bid and offer side).

3. **ADV:** The average daily notional value traded on a particular futures contract.

Bram Kaplan, CFA AC
(1-212) 272-1215
bram.kaplan@jpmorgan.com

J.P.Morgan

Analyst Certification: The Research Analyst(s) denoted by an "AC" on the cover of this report certifies (or, where multiple Research Analysts are primarily responsible for this report, the Research Analyst denoted by an "AC" on the cover or within the document individually certifies, with respect to each security or issuer that the Research Analyst covers in this research) that: (1) all of the views expressed in this report accurately reflect the Research Analyst's personal views about any and all of the subject securities or issuers; and (2) no part of any of the Research Analyst's compensation was, is, or will be directly or indirectly related to the specific recommendations or views expressed by the Research Analyst(s) in this report. For all Korea-based Research Analysts listed on the front cover, if applicable, they also certify, as per KOFIA requirements, that the Research Analyst's analysis was made in good faith and that the views reflect the Research Analyst's own opinion, without undue influence or intervention.

All authors named within this report are Research Analysts who produce independent research unless otherwise specified. In Europe, Sector Specialists (Sales and Trading) may be shown on this report as contacts but are not authors of the report or part of the Research Department.

Important Disclosures

Company-Specific Disclosures: J.P. Morgan does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision. Important disclosures, including price charts and credit opinion history tables (if applicable), are available for compendium reports and all J.P. Morgan–covered companies, and certain non-covered companies, by visiting https://www.jpmm.com/research/disclosures, calling 1-800-477-0406, or e-mailing research.disclosure.inquiries@jpmorgan.com with your request.

Explanation of Equity Research Ratings, Designations and Analyst(s) Coverage Universe:
J.P. Morgan uses the following rating system: Overweight (over the duration of the price target indicated in this report, we expect this stock will outperform the average total return of the stocks in the Research Analyst's, or the Research Analyst's team's, coverage universe); Neutral (over the duration of the price target indicated in this report, we expect this stock will perform in line with the average total return of the stocks in the Research Analyst's, or the Research Analyst's team's, coverage universe); and Underweight (over the duration of the price target indicated in this report, we expect this stock will underperform the average total return of the stocks in the Research Analyst's, or the Research Analyst's team's, coverage universe. NR is Not Rated. In this case, J.P. Morgan has removed the rating and, if applicable, the price target, for this stock because of either a lack of a sufficient fundamental basis or for legal, regulatory or policy reasons. The previous rating and, if applicable, the price target, no longer should be relied upon. An NR designation is not a recommendation or a rating. Some stocks under coverage have a rating but no price target; in these cases, we expect the stock will outperform/perform in line/underperform the average total return of the stocks in the Research Analyst's, or the Research Analyst's team's, coverage universe of the relevant duration of the region. In our Asia (ex-Australia and ex-India) and U.K. small- and mid-cap Equity Research, each stock's expected total return is compared to the expected total return of a benchmark country market index, not to those Research Analysts' coverage universe. If it does not appear in the Important Disclosures section of this report, the certifying Research Analyst's coverage universe can be found on J.P. Morgan's Research website, https://www.jpmorganmarkets.com.

J.P. Morgan Equity Research Ratings Distribution, as of October 03, 2026

	Overweight (buy)	Neutral (hold)	Underweight (sell)
J.P. Morgan Global Equity Research Coverage*	53%	35%	12%
IB clients**	83%	81%	72%
JPMS Equity Research Coverage*	51%	37%	12%
IB clients**	95%	93%	88%

***Please note that the percentages may not add to 100% because of rounding.**
****Percentage of subject companies within each of the "buy," "hold" and "sell" categories for which J.P. Morgan has provided investment banking services within the previous 12 months.**
For purposes of FINRA ratings distribution rules only, our Overweight rating falls into a buy rating category; our Neutral rating falls into a hold rating category; and our Underweight rating falls into a sell rating category. Please note that stocks with an NR designation are not included in the table above. This information is current as of the end of the most recent calendar quarter.

Equity Valuation and Risks: For valuation methodology and risks associated with covered companies or price targets for covered companies, please see the most recent company-specific research report at http://www.jpmorganmarkets.com, contact the primary analyst or your J.P. Morgan representative, or email research.disclosure.inquiries@jpmorgan.com. For material information about the proprietary models used, please see the Summary of Financials in company-specific research reports and the Company Tearsheets, which are available to download on the company pages of our client website, http://www.jpmorganmarkets.com. This report also sets out within it the material underlying assumptions used.

History of Investment Recommendations:
A history of J.P. Morgan investment recommendations disseminated during the preceding 12 months can be accessed on the Research & Commentary page of http://www.jpmorganmarkets.com where you can also search by analyst name, sector or financial instrument.

Analysts' Compensation: The research analysts responsible for the preparation of this report receive compensation based upon various factors, including the quality and accuracy of research, client feedback, competitive factors, and overall firm revenues.

Registration of non-US Analysts: Unless otherwise noted, the non-US analysts listed on the front of this report are employees of non-US affiliates of J.P. Morgan Securities LLC, may not be registered as research analysts under FINRA rules, may not be associated persons of J.P. Morgan Securities LLC, and may not be subject to FINRA Rule 2241 or 2242 restrictions on communications with covered companies, public appearances, and trading securities held by a research analyst account.

Other Disclosures

J.P. Morgan is a marketing name for investment banking businesses of JPMorgan Chase & Co. and its subsidiaries and affiliates worldwide.

UK MIFID FICC research unbundling exemption: UK clients should refer to UK MIFID Research Unbundling exemption for details of J.P. Morgan's implementation of the FICC research exemption and guidance on relevant FICC research categorisation.

All research material made available to clients are simultaneously available on our client website, J.P. Morgan Markets, unless specifically permitted by relevant laws. Not all research content is redistributed, e-mailed or made available to third-party aggregators. For all research material available on a particular stock, please contact your sales representative.

Any long form nomenclature for references to China; Hong Kong; Taiwan; and Macau within this research material are Mainland China; Hong Kong SAR (China); Taiwan (China); and Macau SAR (China).

J.P. Morgan Research may, from time to time, write on issuers or securities targeted by economic or financial sanctions imposed or administered by the governmental authorities of the U.S., EU, UK or other relevant jurisdictions (Sanctioned Securities). Nothing in this report is intended to be read or construed as encouraging, facilitating, promoting or otherwise approving investment or dealing in such Sanctioned Securities. Clients should be aware of their own legal and compliance obligations when making investment decisions.

Digital assets, which may also be referred to as virtual or crypto assets or crypto currencies (e.g. bitcoin), discussed in this research report are subject to a rapidly changing regulatory landscape. The application of relevant laws in respect of such assets may be unsettled, and the legality of any digital asset investment may be unclear. Prior to making any investment decision regarding any digital asset, investors should seek their own legal advice and be aware of any legal considerations regarding the purchase, holding or selling of such an investment. For relevant regulatory advisories on digital assets, including bitcoin and ethereum, please see https://www.jpmorgan.com/disclosures/cryptoasset-disclosure.

The author(s) of this research report may not be licensed to carry on regulated activities in your jurisdiction and, if not licensed, do not hold themselves out as being able to do so.

Exchange-Traded Funds (ETFs): J.P. Morgan Securities LLC ("JPMS") acts as authorized participant for substantially all U.S.-listed ETFs. To the extent that any ETFs are mentioned in this report, JPMS may earn commissions and transaction-based compensation in connection with the distribution of those ETF shares and may earn fees for performing other trade-related services, such as securities lending to short sellers of the ETF shares. JPMS may also perform services for the ETFs themselves, including acting as a broker or dealer to the ETFs. In addition, affiliates of JPMS may perform services for the ETFs, including trust, custodial, administration, lending, index calculation and/or maintenance and other services.

Options and Futures related research: If the information contained herein regards options- or futures-related research, such information is available only to persons who have received the proper options or futures risk disclosure documents. Please contact your J.P. Morgan Representative or visit https://www.theocc.com/components/docs/riskstoc.pdf for a copy of the Option Clearing Corporation's Characteristics and Risks of Standardized Options or https://www.finra.org/sites/default/files/2020-08/Security_Futures_Risk_Disclosure_Statement_2020.pdf for a copy of the Security Futures Risk Disclosure Statement.

Changes to Interbank Offered Rates (IBORs) and other benchmark rates: Certain interest rate benchmarks are, or may in the future become, subject to ongoing international, national and other regulatory guidance, reform and proposals for reform. For more information, please consult: https://www.jpmorgan.com/global/disclosures/interbank_offered_rates

Private Bank Clients: Where you are receiving research as a client of the private banking businesses offered by JPMorgan Chase & Co. and its subsidiaries ("J.P. Morgan Private Bank"), research is provided to you by J.P. Morgan Private Bank and not by any other division of J.P. Morgan, including, but not limited to, the J.P. Morgan Corporate and Investment Bank and its Global Research division.

Legal entity responsible for the production and distribution of research: The legal entity identified below the name of the Reg AC Research Analyst who authored this material is the legal entity responsible for the production of this research. Where multiple Reg AC Research Analysts authored this material with different legal entities identified below their names, these legal entities are jointly responsible for the production of this research. Where more than one legal entity is listed under an analyst's name, the first legal entity is responsible for the production unless stated otherwise. Research Analysts from various J.P. Morgan affiliates may have contributed to the production of this material but may not be licensed to carry out regulated activities in your jurisdiction (and do not hold themselves out as being able to do so). Unless otherwise stated

below in the legal entity disclosures, this material has been distributed by the legal entity responsible for production, or where more than one legal entity is listed under the analyst's name, the first legal entity will be responsible for distribution. If you have any queries, please contact the relevant Research Analyst in your jurisdiction or the entity in your jurisdiction that has distributed this research material.

Legal Entities Disclosures and Country-/Region-Specific Disclosures:
Argentina: JPMorgan Chase Bank N.A Sucursal Buenos Aires is regulated by Banco Central de la República Argentina ("BCRA"- Central Bank of Argentina) and Comisión Nacional de Valores ("CNV"- Argentinian Securities Commission - ALYC y AN Integral N°51).

Australia: J.P. Morgan Securities Australia Limited ("JPMSAL") (ABN 61 003 245 234/AFS Licence No: 238066) is regulated by the Australian Securities and Investments Commission and is a Market Participant of ASX Limited, a Clearing and Settlement Participant of ASX Clear Pty Limited and a Clearing Participant of ASX Clear (Futures) Pty Limited. This material is issued and distributed in Australia by or on behalf of JPMSAL only to "wholesale clients" (as defined in section 761G of the Corporations Act 2001). A list of all financial products covered can be found by visiting https://www.jpmm.com/research/disclosures. J.P. Morgan seeks to cover companies of relevance to the domestic and international investor base across all Global Industry Classification Standard (GICS) sectors, as well as across a range of market capitalisation sizes. If applicable, in the course of conducting public side due diligence on the subject company(ies), the Research Analyst team may at times perform such diligence through corporate engagements such as site visits, discussions with company representatives, management presentations, etc. Research issued by JPMSAL has been prepared in accordance with J.P. Morgan Australia's Research Independence Policy which can be found at the following link: J.P. Morgan Australia - Research Independence Policy.

Brazil: Banco J.P. Morgan S.A. is regulated by the Comissao de Valores Mobiliarios (CVM) and by the Central Bank of Brazil. Ombudsman J.P. Morgan: 0800-7700847 / 0800-7700810 (For Hearing Impaired) / ouvidoria.jp.morgan@jpmchase.com.

Canada: J.P. Morgan Securities Canada Inc. is a registered investment dealer, regulated by the Canadian Investment Regulatory Organization and the Ontario Securities Commission and is the participating member on Canadian exchanges. This material is distributed in Canada by or on behalf of J.P.Morgan Securities Canada Inc.

Chile: Inversiones J.P. Morgan Limitada is an unregulated entity incorporated in Chile.

China: J.P. Morgan Securities (China) Company Limited has been approved by CSRC to conduct the securities investment consultancy business.

Colombia: Banco J.P. Morgan Colombia S.A. is supervised by the Financiera de Colombia (SFC). Any reference in this material to products or services offered abroad by entities other than the Bank in Colombia is included exclusively for descriptive purposes. Such references do not constitute, and should not be construed as, promotional activity or the provision of financial products or services within Colombian territory, as defined under applicable Colombian regulation.

Dubai International Financial Centre (DIFC): JPMorgan Chase Bank, N.A., Dubai Branch is regulated by the Dubai Financial Services Authority (DFSA) and its registered address is Dubai International Financial Centre - The Gate, West Wing, Level 3 and 9 PO Box 506551, Dubai, UAE. This material has been distributed by JP Morgan Chase Bank, N.A., Dubai Branch to persons regarded as professional clients or market counterparties as defined under the DFSA rules.

European Economic Area (EEA): Unless specified to the contrary, research is distributed in the EEA by J.P. Morgan SE ("JPM SE"), which is authorised as a credit institution by the Federal Financial Supervisory Authority (Bundesanstalt für tungsaufsicht, BaFin) and jointly supervised by the BaFin, the German Central Bank (Deutsche Bundesbank) and the European Central Bank (ECB). JPM SE is a company headquartered in Frankfurt with registered address at TaunusTurm, Taunustor 1, Frankfurt am Main, 60310, Germany. The material has been distributed in the EEA to persons regarded as professional investors (or equivalent) pursuant to Art. 4 para. 1 no. 10 and Annex II of MiFID II and its respective implementation in their home jurisdictions ("EEA professional investors"). This material must not be acted on or relied on by persons who are not EEA professional investors. Any investment or investment activity to which this material relates is only available to EEA relevant persons and will be engaged in only with EEA relevant persons.

Hong Kong: J.P. Morgan Securities (Asia Pacific) Limited (CE number AAJ321) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission in Hong Kong, and J.P. Morgan Broking (Hong Kong) Limited (CE number AAB027) is regulated by the Securities and Futures Commission in Hong Kong. JP Morgan Chase Bank, N.A., Hong Kong Branch (CE Number AAL996) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission, is organized under the laws of the United States with limited liability. Where the distribution of this material is a regulated activity in Hong Kong, the material is distributed in Hong Kong by or through J.P. Morgan Securities (Asia Pacific) Limited and/or J.P. Morgan Broking (Hong Kong) Limited.

India: J.P. Morgan India Private Limited (Corporate Identity Number - 68724), having its registered office at J.P. Morgan Tower, Off. C.S.T. Road, Kalina, Santacruz - East, Mumbai – 400098, is registered with the Securities and Exchange Board of India (SEBI) as a 'Research Analyst' having registration number INH000001873. J.P. Morgan India Private Limited is also registered with SEBI as a member of the National Stock Exchange of India Limited and the Bombay Stock Exchange Limited (SEBI Registration Number – INZ000239730) and as a Merchant Banker (SEBI Registration Number - MB/INM000002970). Telephone: 91-22-6157 3000, Facsimile: 91-22-6157 3990 and Website: http://www.jpmipl.com. JPMorgan Chase Bank, N.A. - Mumbai Branch is licensed by the Reserve Bank of India (RBI) (Licence No. 53/ Licence No. BY.4/94; SEBI - IN/CUS/014/ CDSL : IN-DP-CDSL-444-2008/ IN-DP-NSDL-285-2008/ INBI00000984/ INE231311239) as a Scheduled Commercial Bank in India, which is its primary license allowing it to carry on Banking business in India and other activities, which a Bank branch in India are permitted to undertake. For non-local research material, this material is not distributed in India by J.P. Morgan India

Private Limited. Compliance Officer: Ashutosh Sharma; ashutosh.j.sharma@jpmchase.com; +912261575002. Grievance Officer: Ramprasadh K, jpmipl.research.feedback@jpmorgan.com; +912261573000. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Please visit Terms and Conditions and Most Important Terms and Conditions (MITC). The annual Compliance audit report is available at http://www.jpmipl.com/#research.

Indonesia: PT J.P. Morgan Sekuritas Indonesia is a member of the Indonesia Stock Exchange and is registered and supervised by the Otoritas Jasa Keuangan (OJK).

Korea: J.P. Morgan Securities (Far East) Limited, Seoul Branch, is a member of the Korea Exchange (KRX). JPMorgan Chase Bank, N.A., Seoul Branch, is licensed as a branch office of foreign bank (JPMorgan Chase Bank, N.A.) in Korea. Both entities are regulated by the Financial Services Commission (FSC) and the Financial Supervisory Service (FSS). For non-macro research material, the material is distributed in Korea by or through J.P. Morgan Securities (Far East) Limited, Seoul Branch.

Japan: JPMorgan Securities Japan Co., Ltd. and JPMorgan Chase Bank, N.A., Tokyo Branch are regulated by the Financial Services Agency in Japan.

Malaysia: This material is issued and distributed in Malaysia by JPMorgan Securities (Malaysia) Sdn Bhd (18146-X), which is a Participating Organization of Bursa Malaysia Berhad and holds a Capital Markets Services License issued by the Securities Commission in Malaysia.

Mexico: J.P. Morgan Casa de Bolsa, S.A. de C.V., J.P. Morgan Grupo Financiero is member of the Mexican Stock Exchange ("Bolsa Mexicana de Valores") and the Institutional Stock Exchange ("Bolsa Institucional de Valores"), and it is authorized to act as a broker dealer by the National Banking and Securities Exchange Commission ("Comisión Nacional Bancaria y de Valores").

New Zealand: This material is issued and distributed by JPMSAL in New Zealand only to "wholesale clients" (as defined in the Financial Markets Conduct Act 2013). JPMSAL is registered as a Financial Service Provider under the Financial Service providers (Registration and Dispute Resolution) Act of 2008.

Nigeria: J.P. Morgan Chase Representative Office (Nigeria) Limited is regulated by the Central Bank of Nigeria.

Philippines: J.P. Morgan Securities Philippines Inc. is a Trading Participant of the Philippine Stock Exchange and a member of the Securities Clearing Corporation of the Philippines and the Securities Investor Protection Fund. It is regulated by the Securities and Exchange Commission.

Singapore: This material is issued and distributed in Singapore by or through J.P. Morgan Securities Singapore Private Limited (JPMSS) [MDDI (P) 070/08/2026 and Co. Reg. No.: 199405335R], which is a member of the Singapore Exchange Securities Trading Limited, and/or JPMorgan Chase Bank, N.A., Singapore branch (JPMCB Singapore), both of which are regulated by the Monetary Authority of Singapore. This material is issued and distributed in Singapore only to accredited investors, expert investors and institutional investors, as defined in Section 4A of the Securities and Futures Act, Cap. 289 (SFA). This material is not intended to be issued or distributed to any retail investors or any other investors that do not fall into the classes of "accredited investors," "expert investors" or "institutional investors," as defined under Section 4A of the SFA. Recipients of this material in Singapore are to contact JPMSS or JPMCB Singapore in respect of any matters arising from, or in connection with, the material.

South Africa: J.P. Morgan Equities South Africa Proprietary Limited and JPMorgan Chase Bank, N.A., Johannesburg Branch are members of the Johannesburg Securities Exchange and are regulated by the Financial Services Conduct Authority (FSCA).

Taiwan: J.P. Morgan Securities (Taiwan) Limited is a participant of the Taiwan Stock Exchange (company-type) and regulated by the Taiwan Securities and Futures Bureau. Material relating to equity securities is issued and distributed in Taiwan by J.P. Morgan Securities (Taiwan) Limited, subject to the license scope and the applicable laws and the regulations in Taiwan. **To the extent that J.P. Morgan Securities (Taiwan) Limited produces research materials on securities not listed on the Taiwan Stock Exchange or Taipei Exchange ("Non-Taiwan Listed Securities"), these materials shall not constitute securities recommendations for the purpose of applicable Taiwan regulations, and, for the avoidance of doubt, J.P. Morgan Securities (Taiwan) Limited does not act as broker for Non-Taiwan Listed Securities.** According to Paragraph 2, Article 7-1 of Operational Regulations Governing Securities Firms Recommending Trades in Securities to Customers (as amended or supplemented) and/or other applicable laws or regulations, please note that the recipient of this material is not permitted to engage in any activities in connection with the material that may give rise to conflicts of interests, unless otherwise disclosed in the "Important Disclosures" in this material.

Thailand: This material is issued and distributed in Thailand by JPMorgan Securities (Thailand) Ltd., which is a member of the Stock Exchange of Thailand and is regulated by the Ministry of Finance and the Securities and Exchange Commission. The registered address is 548 One City Center Building, 50th Floor, Ploenchit Road, Lumphini, Pathum Wan, Bangkok 10330.

UK: Research is produced in the UK by J.P. Morgan Securities plc ("JPMS plc") which is a member of the London Stock Exchange and is authorised by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority or J.P. Morgan Markets Limited ("JPMML Ltd") which is authorised and regulated by the Financial Conduct Authority. Unless specified to the contrary, this material is distributed in the UK by JPMS plc and is directed in the UK only to: (a) persons having professional experience in matters relating to investments falling within article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) (Order) 2005 ("the FPO"); (b) persons outlined in article 49 of the FPO (high net worth companies, unincorporated associations or partnerships, the trustees of high value trusts, etc.); or (c) any persons to whom this communication may otherwise lawfully be made; all such persons being referred to as "UK relevant persons". This material must not be acted on or relied on by persons who are not UK relevant persons. Any investment or

investment activity to which this material relates is only available to UK relevant persons and will be engaged in only with UK relevant persons. A description of J.P. Morgan EMEA's policy for prevention and avoidance of conflicts of interest related to the production of Research can be found at the following link: J.P. Morgan EMEA - Research Independence Policy.

U.S.: J.P. Morgan Securities LLC ("JPMS") is a member of the NYSE, FINRA, SIPC, and the NFA. JPMorgan Chase Bank, N.A. is a member of the FDIC. Material published by non-U.S. affiliates is distributed in the U.S. by JPMS who accepts responsibility for its content.

General: Additional information is available upon request. The information in this material has been obtained from sources believed to be reliable. While all reasonable care has been taken to ensure that the facts stated in this material are accurate and that the forecasts, opinions and expectations contained herein are fair and reasonable, JPMorgan Chase & Co. or its affiliates and/or subsidiaries (collectively J.P. Morgan) make no representations or warranties whatsoever to the completeness or accuracy of the material provided, except with respect to any disclosures relative to J.P. Morgan and the Research Analyst's involvement with the issuer that is the subject of the material. Accordingly, no reliance should be placed on the accuracy, fairness or completeness of the information contained in this material. There may be certain discrepancies with data and/or limited content in this material as a result of calculations, adjustments, translations to different languages, and/or local regulatory restrictions, as applicable. These discrepancies should not impact the overall investment analysis, views and/or recommendations of the subject company(ies) that may be discussed in the material. Artificial intelligence tools may have been used in the preparation of this material, including assisting in data analysis, pattern recognition, and content drafting for research material. J.P. Morgan accepts no liability whatsoever for any loss arising from any use of this material or its contents, and neither J.P. Morgan nor any of its respective directors, officers or employees, shall be in any way responsible for the contents hereof, apart from the liabilities and that may be imposed on them by the relevant regulatory authority in the jurisdiction in question, or the regulatory regime thereunder. Opinions, forecasts or projections contained in this material represent J.P. Morgan's current opinions or judgment as of the date of the material only and are therefore subject to change without notice. Periodic updates may be provided on companies/industries based on company-specific developments or announcements, market conditions or any other publicly available information. There can be no assurance that future results or events will be consistent with any such opinions, forecasts or projections, which represent only one possible outcome. Furthermore, such opinions, forecasts or projections are subject to certain risks, uncertainties and assumptions that have not been verified, and future actual results or events could differ materially. The value of, or income from, any investments referred to in this material may fluctuate and/or be affected by changes in exchange rates. All pricing is indicative as of the close of market for the securities discussed, unless otherwise stated. Past performance is not indicative of future results. Accordingly, investors may receive back less than originally invested. This material is not intended as an offer or solicitation for the purchase or sale of any financial instrument. The opinions and recommendations herein do not take into account individual client circumstances, objectives, or needs and are not intended as recommendations of particular securities, financial instruments or strategies to particular clients. This material may include views on structured securities, options, futures and other derivatives. These are complex instruments, may involve a high degree of risk and may be appropriate investments only for sophisticated investors who are capable of understanding and assuming the risks involved. The recipients of this material must make their own independent decisions regarding any securities or financial instruments mentioned herein and should seek advice from such independent financial, legal, tax or other adviser as they deem necessary. J.P. Morgan may trade as a principal on the basis of the Research Analysts' views and research, and it may also engage in transactions for its own account or for its clients' accounts in a manner inconsistent with the views taken in this material, and J.P. Morgan is under no obligation to ensure that such other communication is brought to the attention of any recipient of this material. Others within J.P. Morgan, including Strategists, Sales staff and other Research Analysts, may take views that are inconsistent with those taken in this material. Employees of J.P. Morgan not involved in the preparation of this material may have investments in the securities (or derivatives of such securities) mentioned in this material and may trade them in ways different from those discussed in this material. This material is not an advertisement for or marketing of any issuer, its products or services, or its securities in any jurisdiction.

Confidentiality and Security Notice: This transmission may contain information that is privileged, confidential, legally privileged, and/or exempt from disclosure under applicable law. If you are not the intended recipient, you are hereby notified that any disclosure, copying, distribution, or use of the information contained herein (including any reliance thereon) is STRICTLY PROHIBITED. Although this transmission and any attachments are believed to be free of any virus or other defect that might affect any computer system into which it is received and opened, it is the responsibility of the recipient to ensure that it is virus free and no responsibility is accepted by JPMorgan Chase & Co., its subsidiaries and affiliates, as applicable, for any loss or damage arising in any way from its use. If you received this transmission in error, please immediately contact the sender and destroy the material in its entirety, whether in electronic or hard copy format. This message is subject to electronic monitoring: https://www.jpmorgan.com/disclosures/email

MSCI: Certain information herein ("Information") is reproduced by permission of MSCI Inc., its affiliates and information providers ("MSCI") ©2026. No reproduction or dissemination of the Information is permitted without an appropriate license. MSCI MAKES NO EXPRESS OR IMPLIED WARRANTIES (INCLUDING MERCHANTABILITY OR FITNESS) AS TO THE INFORMATION AND DISCLAIMS ALL LIABILITY TO THE EXTENT PERMITTED BY LAW. No Information constitutes investment advice, except for any applicable Information from MSCI ESG Research. Subject also to msci.com/disclaimer

Any mentions of third-party trademarks, brand names, products, and services are for referential purposes only and any mention thereof is not meant to imply any sponsorship, endorsement, or affiliation.

Sustainalytics: Certain information, data, analyses and opinions contained herein are reproduced by permission of Sustainalytics and: (1) includes the proprietary information of Sustainalytics; (2) may not be copied or redistributed except as specifically authorized; (3) do not constitute investment advice nor an endorsement of any product or project; (4) are provided solely for informational purposes; and (5) are not warranted to be complete, accurate or timely. Sustainalytics is not responsible for any trading decisions, damages or other losses related to it or

its use. The use of the data is subject to conditions available at https://www.sustainalytics.com/legal-disclaimers. ©2026 Sustainalytics. All Rights Reserved.

"Other Disclosures" last revised October 17, 2026.

Copyright 2026 JPMorgan Chase & Co. All rights reserved. This material or any portion hereof may not be reprinted, sold or redistributed without the written consent of J.P. Morgan. Without prior written consent from J.P. Morgan, it is prohibited to (A) use or share any material, commentary, information or other data received from J.P. Morgan or an authorized third-party ("J.P. Morgan Data") in any artificial intelligence ("AI") systems or models, or (B) use automated means, including robots, crawlers, web-scrapers or other automated tools or scripts, to access, collect, extract or harvest any J.P. Morgan Data, including scraping of user data, content, metadata, pricing information or other information.

Completed 26 Oct 2026 03:44 PM EDT

Disseminated 26 Oct 2026 03:44 PM EDT